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Analog peak power management for multi-die operations

Abstract

Methods of peak power management (PPM) for a storage system having multiple memory dies are disclosed. Each memory die includes a first PPM circuit and a second PPM circuit. First PPM circuits of the multiple memory dies are electrically connected to form a first PPM group. Similarly, second PPM circuits are electrically connected to form a second PPM group. Peak power operations can be managed by switching on a first pull-down driver of the first PPM circuit on a selected memory die when a first PPM enablement signal of the first PPM group is zero; waiting for a first delay period; switching on a second pull-down driver of the second PPM circuit on the selected memory die when a second PPM enablement signal of the second PPM group is zero. The PPM enablement signals depend on the current flowing through each pull-down driver in the first and second PPM groups.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND INCORPORATION BY REFERENCE (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/116,253 filed on Dec. 9, 2020 and titled “Analog Peak Power Management for Multi-die Operations,” which claims priority to PCT/CN2020/128024 filed on Nov. 11, 2020, all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to the field of semiconductor technology, and more particularly, to circuit designs and methods for peak power management in a storage system.

BACKGROUND

(2) In many servers and mobile devices, NAND storage system is widely used as the primary non-volatile storage device due to its high storage density and relatively low access latency. However, performance of a high density storage system, for example, a three-dimensional (3D) NAND storage system, is often restricted by the maximum amount of power (or peak current) that it can use. Currently, operations consuming high power (i.e., peak power operations) that are carried out by various memory dies of the NAND storage system, can be staggered by a system controller. Only a limited number of peak power operations can be performed simultaneously. This approach can also result in increased system loading with unnecessary over-management. Communications between different memory dies can be established to coordinate the peak power operations. Currently, coordination between two memory dies can be arranged and peak power operations can be staggered between these two memory dies. When memory dies are grouped into two dies per group, coordination between groups remains an issue. Other approaches can provide multiple memory dies in each group to coordinate peak power operations, but are also limited to one peak power operation in each group. Therefore, it is necessary to optimize the control circuits and peak power management scheme to coordinate multiple memory dies simultaneously to allow multiple peak power operations performed on a storage system. As such, the storage system's power or current budget can be fully utilized.

BRIEF SUMMARY

(3) The present disclosure is directed to providing effective peak power management for a memory storage system.

(4) One aspect of the present disclosure provides a peak power management (PPM) system for a storage system with multiple memory dies. The PPM system includes two or more PPM groups, each of the two or more PPM groups having multiple PPM circuits. Each of the multiple PPM circuits includes a pull-up driver electrically connected to a power source and a pull-up resistor; a pull-down driver electrically connected to a pull-down resistor; and a PPM pin connected to the pull-up resistor and the pull-down resistor. The PPM pins in each of the two or more PPM groups are electrically connected with each other. Each of the two or more PPM groups is configured to manage m number of peak power operations based on an electric potential of the PPM pins, where m is a whole number.

(5) In some implementations, each memory die includes at least one PPM circuit.

(6) In some implementations, the electric potentials of the PPM pins in each of the two or more PPM groups is determined by pull-down currents flowing through the pull-down drivers in the multiple PPM circuits.

(7) In some implementations, each of the two or more PPM groups further includes a comparator with a first input terminal electrically connected to the PPM pins and a second input terminal electrically connected to a reference voltage. An output terminal of the comparator is connected to an inverter.

(8) In some implementations, each of the two or more PPM groups further includes a resistance/capacitance (RC) filter electrically connected to the PPM pins and the first input terminal of the comparator.

(9) In some implementations, the reference voltage is in a range between a first electric potential of the PPM pins and a second electric potential of the PPM pins, the first electric potential and the second electric potential of the PPM pins corresponding to $m-1$ and m number of peak power operations, respectively.

(10) In some implementations, the first electric potential of the PPM pins is

$$(11) \frac{R_d}{R_d + (m-1) \cdot R_u} V_{dd},$$

and the second electric potential of the PPM pins is

$$(12) \frac{R_d}{R_d + m \cdot R_u} V_{dd},$$

wherein R_u is the resistance of the pull-up resistor; R_d is the resistance of the pull-down resistor; and V_{dd} is the power source voltage.

(13) In some implementations, the pull-up driver is a p-channel metal oxide semiconductor field effect transistor (MOSFET), and the pull-down driver is an n-channel metal oxide semiconductor field effect transistor (MOSFET).

(14) In some implementations, the PPM pins in each PPM group are electrically connected through die-to-die connections, each die-to-die connection comprising a metal interconnect.

(15) In some implementations, the PPM pins in each PPM group are electrically connected through flip-chip bonding, die-to-die bonding, or wire-bonding.

(16) The present disclosure also provides a method of peak power management (PPM) for a storage system with multiple memory dies. Each of the multiple memory dies includes a first PPM circuit and a second PPM circuit. First PPM circuits of the multiple memory dies are electrically connected to form a first PPM group, and second PPM circuits of the multiple memory dies are electrically connected to form a second PPM group. The method includes the following steps: switching on a first pull-down driver of the first PPM circuit on a selected memory die when a first PPM enablement signal of the first PPM group is zero; waiting for a first delay period; switching on a second pull-down driver of the second PPM circuit on the selected memory die when a second PPM enablement signal of the second PPM group is zero; and performing a peak power operation on the selected memory die. The first and second PPM enablement signals depend on the current flowing through each of the pull-down drivers in the first and second PPM groups.

(17) In some implementations, after performing the peak power operation, the second pull-down driver of the second PPM circuit on the selected memory die is switched off.

(18) In some implementations, after switching off the second pull-down driver of the second PPM circuit, the first pull-down driver of the first PPM circuit is switched off.

(19) In some implementations, prior to switching off the first pull-down driver of the first PPM circuit, a second delay period is performed.

(20) In some implementations, the second delay period is a predetermined time period.

(21) In some implementations, the first delay period is a time period unique to the selected memory die among the multiple memory dies on the storage system.

(22) In some implementations, the method further includes polling and checking, repeatedly, the first PPM enablement signal and the second PPM enablement signal when the first PPM enablement signal and/or the second PPM enablement signal is not zero.

(23) In some implementations, the method further includes generating the first PPM enablement signal by comparing a reference voltage with a first electric potential of a first plurality of PPM pins of the first PPM circuits in the first PPM group. The first PPM circuits are electrically connected through the first plurality of PPM pins. The second PPM is configured to generate an enablement signal by comparing the reference voltage with a second electric potential of a second plurality of PPM pins of the second PPM circuits in the second PPM group, wherein the second PPM circuits are electrically connected through the second plurality of PPM pins.

(24) In some implementations, the generating of the first PPM enablement signal further includes setting the first PPM enablement signal to 0 if the first electric potential of the first plurality of

PPM pins is higher than the reference voltage. The second PPM is configured to generate an enablement signal further comprises setting the second PPM enablement signal to 0 if the second electric potential of the second plurality of PPM pins is higher than the reference voltage.

(25) In some implementations, the generating of the first PPM enablement signal further comprises setting the first PPM enablement signal to 1 if the first electric potential of the first plurality of PPM pins is less than the reference voltage. The second PPM is configured to generate an enablement signal further comprises setting the second PPM enablement signal to 1 if the second electric potential of the second plurality of PPM pins is less than the reference voltage.

(26) In some implementations, the method also includes selecting the reference voltage based on a maximum m number of peak power operations for the storage system.

(27) In some implementations, the method further includes regulating the first electric potential of the first plurality of PPM pins through the first pull-down driver; and regulating the second electric potential of the second plurality of PPM pins through the second pull-down driver.

(28) Another aspect of the present disclosure provides a peak power management (PPM) circuit for a storage system with multiple memory dies. The PPM circuit includes a pull-up driver electrically connected between a power source and a PPM pin. A pull-down driver is electrically connected between a ground and the PPM pin, wherein the pull-up driver and the pull-down driver are connected in series. The PPM circuit is configured to manage a peak power operation for the memory die based on an electric potential of the PPM pin, wherein PPM pins of PPM circuits on different memory dies are electrically connected.

(29) In some implementations, the PPM circuit further includes a pull-up resistor electrically connected between the power source and the PPM pin, wherein the pull-up resistor is connected in series with the pull-up driver.

(30) In some implementations, the PPM circuit further includes a pull-down resistor electrically connected between the ground and the PPM pin, wherein the pull-down resistor is connected in series with the pull-down driver.

(31) In some implementations, the PPM circuit further includes a pull-up current source electrically connected between the power source and the PPM pin, wherein the pull-up current source is connected in series with the pull-up driver and is configured to provide constant current.

(32) In some implementations, the PPM circuit further includes a pull-down current source electrically connected between the ground and the PPM pin, wherein the pull-down current source is connected in series with the pull-down driver and is configured to provide constant current.

(33) In some implementations, the PPM circuit further includes a comparator with a first input terminal electrically connected to the PPM pin and a second input terminal electrically connected to a reference voltage. The reference voltage is based on a maximum number of peak power operations allowed in the storage system. In some implementations, the PPM circuit further includes an inverter with an input connected to an output terminal of the comparator.

(34) Yet another aspect of the present disclosure discloses a method of peak power management (PPM) for a storage system with multiple memory dies. Each of the multiple memory dies comprises a PPM circuit, and each PPM circuit comprises a pull-up driver connected in series with a pull-down driver. The method includes sending a first control signal to switch on a corresponding pull-up driver on one of the multiple memory dies; sending a second control signal to switch on a respective pull-down driver on a selected memory die based on a PPM enablement signal; and performing a peak power operation on the selected memory die.

(35) In some implementations, the sending of the second control signal includes applying an analog voltage on a gate terminal of the respective pull-down driver.

(36) In some implementations, the method further includes generating a pull-down current proportional to a peak power current for the peak power operation.

(37) In some implementations, the method further includes generating the PPM enablement signal by comparing a reference voltage with an electric potential of a PPM pin of the PPM circuit,

wherein PPM pins of PPM circuits on different memory dies are electrically connected.

(38) In some implementations, the method further includes selecting the reference voltage based on a maximum number of peak power operations for the storage system. In some implementations, the method further includes generating, by a pull-down current source, a pull-down current that is proportional to a peak power current for the peak power operation.

(39) The present disclosure further discloses a method of peak power management (PPM) for a storage system with multiple memory dies, wherein each of the multiple memory dies comprises a PPM circuit, and each PPM circuit comprises a pull-up driver connected in series with a pull-down driver. The method includes sending a second control signal to switch on a corresponding pull-down driver on one of the multiple memory dies; sending a first control signal to switch on a respective pull-up driver on a selected memory die based on a PPM enablement signal; and performing a peak power operation on the selected memory die.

(40) In some implementations, the sending of the first control signal comprises applying an analog voltage on a gate terminal of the respective pull-up driver.

(41) In some implementations, the method further includes generating a pull-up current proportional to a peak power current for the peak power operation. In some implementations, the method further includes generating the PPM enablement signal by comparing a reference voltage with an electric potential of a PPM pin of the PPM circuit, wherein PPM pins of PPM circuits on different memory dies are electrically connected.

(42) In some implementations, the method further includes selecting the reference voltage based on a maximum number of peak power operations for the storage system.

(43) In some implementations, the method further includes generating, by a pull-up current source, a pull-up current that is proportional to a peak power current for the peak power operation.

(44) Other aspects of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate embodiments of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

(2) FIG. 1A illustrates a storage system with one or more NAND flash memories, according to some embodiments of the present disclosure.

(3) FIG. 1B illustrates a top-down view of a memory die, according to some embodiments of the present disclosure.

(4) FIGS. 1C and 1D illustrate exemplary storage systems with one or more NAND flash memories, according to some embodiments of the present disclosure.

(5) FIG. 2 illustrates a peak power management system in a storage system, according to some embodiments of the present disclosure.

(6) FIG. 3 illustrates a peak power management circuit, according to some embodiments of the present disclosure.

(7) FIG. 4 illustrates an equivalent circuit of a peak power management group, according to some embodiments of the present disclosure.

(8) FIG. 5 illustrates a peak power management scheme, according to some embodiments of the present disclosure.

(9) FIG. 6 illustrates a peak power check routine, according to some embodiments of the present disclosure.

- (10) FIG. 7 illustrates a peak power management circuit, according to some embodiments of the present disclosure.
- (11) FIG. 8 illustrates an equivalent circuit of a peak power management group, according to some embodiments of the present disclosure.
- (12) FIG. 9 illustrates a peak power management scheme, according to some embodiments of the present disclosure.
- (13) FIG. 10 illustrates a peak power check routine, according to some embodiments of the present disclosure.
- (14) FIG. 11 illustrates a peak power management circuit, according to some embodiments of the present disclosure.
- (15) FIG. 12 illustrates an equivalent circuit of a peak power management group, according to some embodiments of the present disclosure.
- (16) FIG. 13 illustrates a peak power management scheme, according to some embodiments of the present disclosure.
- (17) FIG. 14 illustrates a peak power management circuit, according to some embodiments of the present disclosure.
- (18) FIG. 15 illustrates an equivalent circuit of a peak power management group, according to some embodiments of the present disclosure.
- (19) FIG. 16 illustrates a peak power management scheme, according to some embodiments of the present disclosure.
- (20) The features and advantages of the present invention will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.
- (21) Embodiments of the present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

- (22) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. A person skilled in the pertinent art will recognize that other configurations and arrangements can be used without departing from the spirit and scope of the present disclosure. It will be apparent to a person skilled in the pertinent art that the present disclosure can also be employed in a variety of other applications.
- (23) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “some embodiments,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of a person skilled in the pertinent art to affect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.
- (24) In general, terminology can be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, can be used to describe any feature, structure, or characteristic in a singular sense or can be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, can be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” can be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.
- (25) As used herein, the term “nominal/nominally” refers to a desired, or target, value of a

characteristic or parameter for a component or a process step, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the term “about” indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the term “about” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., $\pm 10\%$, $\pm 20\%$, or $\pm 30\%$ of the value).

(26) FIG. 1A illustrates a block diagram of an exemplary system **S1** having a storage system **10**, according to some embodiments of the present disclosure. System **S1** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. The storage system **10** (also referred to as a NAND storage system) can include a memory controller **20** and one or more NAND flash memories **100** (e.g., **100-1**, **100-2**, **100-3**, . . . , **100-n**). The storage system **10** can communicate with a host computer **15** through the memory controller **20**, where the memory controller **20** can be connected to the one or more NAND flash memories **100-1**, **100-2**, **100-3**, . . . , **100-n**, via one or more memory channels **30** (e.g., **30-1**, **30-2**, **30-3**, . . . , **30-n**). In some embodiments, each NAND flash memory **100** can be managed by the memory controller **20** via a memory channel **30**.

(27) In some embodiments, the host computer **15** can include a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). The host computer **15** sends data to be stored at the storage system **10** or retrieves data by reading the storage system **10**. The memory controller **20** can handle I/O requests received from the host computer **15**, ensure data integrity and efficient storage, and manage the NAND flash memory **100**. The memory channels **30** can provide data and control communication between the memory controller **20** and each NAND flash memory **100** via a data bus. The memory controller **20** can select one of the NAND flash memory **100** according to a chip enable signal.

(28) FIG. 1B illustrates a top-down view of a NAND flash memory **100**, according to some embodiments of the present disclosure. The NAND flash memory **100** can be a memory die (or a die) or any portion of a memory die. In some embodiments, each NAND flash memory **100** can include one or more memory planes **101**, each of which can include a plurality of memory blocks **103**. Identical and concurrent operations can take place at each memory plane **101**. The memory block **103**, which can be megabytes (MB) in size, is the smallest size to carry out erase operations. Shown in FIG. 1B, the exemplary NAND flash memory **100** includes four memory planes **101** and each memory plane **101** includes six memory blocks **103**. Each memory block **103** can include a plurality of memory cells, where each memory cell can be addressed through interconnections such as bit lines and word lines. The bit lines and word lines can be laid out perpendicularly (e.g., in rows and columns, respectively), forming an array of metal lines. The direction of bit lines and word lines are labeled as “BL” and “WL” in FIG. 1B. In this disclosure, the memory block **103** is also referred to as the “memory array” or “array.” The memory array is the core area on a memory die, performing storage functions.

(29) The NAND flash memory **100** also includes a peripheral region **105**, an area surrounding memory planes **101**. The peripheral region **105** contains periphery circuits, which include many digital, analog, and/or mixed-signal circuits to support functions of the memory array, for example, page buffers/sense amplifiers **50**, row decoders **40**, column decoders **60**, and control circuits **70**. Control circuits **70** include MCU, state machine, register, active and/or passive semiconductor devices, such as transistors, diodes, capacitors, resistors, etc. The control circuits **70** of the peripheral region **105** can be configured to initiate a program operation on a select memory cell of a NAND memory string in the memory block **103**. In some implementations, the control circuits **70** can receive a command (e.g., a program/read/erase command) from a memory controller (e.g., the

memory controller **20**) through a communication interface of the NAND flash memory **100**, and in response, sends control signals to at least row decoder/word line driver, column decoder/bit line driver, and voltage generator disposed in the peripheral region **105** to initiate the program/read/erase operation on the select memory cell.

(30) It is noted that the layout of the electronic components in the storage system **10** and the NAND flash memory **100** in FIGS. **1A** and **1B** are shown as examples. The storage system **10** and the NAND flash memory **100** can have other layout and can include additional components. For example, the NAND flash memory **100** can also have high-voltage charge pumps, I/O circuits, etc. The storage system **10** can also include firmware, data scrambler, etc.

(31) Memory controller **20** and one or more NAND flash memory **100** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an eMMC package. That is, storage system **10** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. **1C**, memory controller **20** and a single NAND flash memory **100** can be integrated into a memory card **26**. Memory card **26** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc. Memory card **26** can further include a memory card connector **24** coupling memory card **26** with a host (e.g., the host computer **15** in FIG. **1A**). In another example as shown in FIG. **1D**, memory controller **20** and multiple NAND flash memories **100** can be integrated into a solid state drive (SSD) **27**. SSD **27** can further include an SSD connector **28** coupling SSD **27** with a host (e.g., the host computer **15** in FIG. **1A**).

(32) FIG. **2** illustrates a peak power management system **200**, according to some embodiments of the present disclosure. The peak power management (PPM) system **200** can be implemented in the storage system **10** in FIG. **1A**, having a plurality of memory dies **100-1**, **100-2**, **100-3**, . . . , **100-n**, and each memory die can be similar to the NAND flash memory **100** discussed previously in reference with FIG. **1B**. In some embodiments, each NAND flash memory **100** can include at least one peak power management (PPM) circuit. In the example shown in FIG. **2**, each NAND flash memory **100** includes a first PPM (PPM1) circuit **202** and a second PPM (PPM2) circuit **206**. Each PPM1 circuit and each PPM2 circuit can include a PPM contact pad (also referred to as PPM pin). For example, each PPM1 circuit **202** includes a PPM1 pin **204** (also referred to as PPM1 pin), and each PPM2 circuit **206** includes a PPM2 pin **208** (also referred to as PPM2 pin). The PPM1 circuits **202-1**, **202-2**, **202-3**, . . . , **202-n** on different NAND flash memories **100-1**, **100-2**, **100-3**, . . . , **100-n** can communicate with each other through the PPM1 pins **204-1**, **204-2**, **204-3**, . . . , **204-n**. The PPM2 circuits **206-1**, **206-2**, **206-3**, . . . , **206-n** on different NAND flash memories **100-1**, **100-2**, **100-3**, . . . , **100-n** can communicate with each other through the PPM2 pins **208-1**, **208-2**, **208-3**, . . . , **208-n**. In some embodiments, the PPM pins between different NAND flash memories **100** can be electrically connected with each other through a plurality of die-to-die connections. For example, the PPM1 pins **204** between different NAND flash memories **100** can be electrically connected with each other through first die-to-die connections **205**, and are thereby also referred to as a first PPM group **210** (or PPM1 group). Likewise, the PPM2 pins **208** between different NAND flash memories **100** can be electrically connected with each other through second die-to-die connections **209**, and are thereby also referred to as a second PPM group **211** (or PPM2 group). In some embodiments, the first and second die-to-die connections **205** and **209** can be a metal wire formed through wire-bonding. In some embodiments, the first and second die-to-die connections **205** and **209** can be metal wires or any suitable metal or conductive material formed through flip-chip bonding or any suitable die-to-die bonding. In some embodiments, the first and second die-to-die connections **205** and **209** can be formed by through-silicon VIAs (e.g., through-array structures).

(33) By using the die-to-die connections described above, communications between different memory dies (i.e., NAND flash memories **100-1**, **100-2**, **100-3**, . . . , **100-n**) can be established in

the storage system. As such, the storage system **10** can send operation commands to any number of memory dies at any time while the PPM1 circuits **202** and the PPM2 circuits **206** can control the system's power consumption by selecting one or more memory dies. In one example, each memory die shown in FIGS. **1A-1B** and **2** can have the communication interface, where commands from the memory controller **20** can be received by the control circuit **70** of each memory die. In some embodiments, the first PPM (PPM1) circuit **202** and the second PPM (PPM2) circuit **206** can be implemented in the peripheral region **105** on a memory die, where the control circuits **70** of the memory die can send control signals to the first PPM (PPM1) circuit **202** and the second PPM (PPM2) circuit **206** on the memory die in response to the commands received at the communication interface. In some embodiments, the commands received at the communication interface include a PPM circuit selecting signal for selecting the PPM1 circuit **202** and/or the PPM2 circuit **206** on a particular memory die.

(34) FIG. **3** illustrates an exemplary PPM circuit **300** provided in each of the NAND flash memories **100**, according to some embodiments of the present disclosure, where each of the PPM1 circuits **202** and each of the PPM2 circuits **206** shown in FIG. **2** can be similar to the PPM circuit **300**. The PPM circuit **300** can include a pull-up driver **314**, where one terminal of the pull-up driver **314** is connected to a power source **312** with a voltage $V_{sub.dd}$. In some embodiments, the power source **312** can be any suitable power supply that has desired noise sensitivity. In some embodiments, the pull-up driver **314** can be a transistor, for example, a metal-oxide-semiconductor-field-effect-transistor (MOSFET). In some embodiments, the pull-up driver **314** can be a p-channel MOSFET (i.e., pFET), where a source terminal of the pull-up driver **314** can be connected to the power source **312** and a drain terminal of the pull-up driver **314** can be connected to a pull-up resistor **318** with a resistance $R_{sub.u}$.

(35) In some embodiments, the PPM circuit **300** also includes a pull-down driver **336**. In some embodiments, the pull-down driver **336** can be a transistor, for example, a MOSFET. In some embodiments, the pull-down driver **336** can be an n-channel MOSFET (i.e., nFET). A source terminal of the pull-down driver **336** can be grounded, and a drain terminal of the pull-down driver **336** can be connected to a pull-down resistor **320** with a resistance $R_{sub.d}$.

(36) In some embodiments, the pull-up resistor **318** and the pull-down resistor **320** are electrically connected in series at a node **322**, where the PPM contact pad (e.g., the PPM1 pin **204** or the PPM2 pin **208** in FIG. **2**) can be electrically connected to the node **322**. As described previously, all the PPM contact pads (e.g., the PPM1 pins **204**) in one PPM group (e.g., the PPM1 group **210**) can be electrically connected through the die-to-die connections (e.g., the first die-to-die connection **205**). Thus, all the PPM contact pads of the same PPM group can be held to an electrical potential $V_{sub.ppm}$ at the node **322**.

(37) In some embodiments, the PPM circuit **300** can also include an output circuit including a comparator **328**, with a first input terminal **324** at a reference voltage $V_{sub.ref}$ and a second input terminal **326** connected to the node **322**. The comparator **328** can be an operational amplifier used for comparing an input voltage $V_{sub.in}$ at the second input terminal **326** with the reference voltage $V_{sub.ref}$ at the first input terminal **324**, where an output voltage $V_{sub.out}$ at an output terminal **330** can indicate whether the input voltage $V_{sub.in}$ is above or below the reference voltage $V_{sub.ref}$. For example, the output voltage $V_{sub.out}$ can be a positive voltage when the input voltage $V_{sub.in}$ is larger than the reference voltage $V_{sub.ref}$. On the other hand, the output voltage $V_{sub.out}$ can be a negative voltage when the input voltage $V_{sub.in}$ is smaller than the reference voltage $V_{sub.ref}$. In some embodiments, the comparator **328** can be an analog comparator.

(38) In some embodiments, the output circuit of the PPM circuit **300** can further include an inverter **332** with an input terminal connected to the output terminal **330** of the comparator **328**. The inverter **332** can invert an input signal. For example, when the output voltage $V_{sub.out}$ of the comparator **328** is a positive voltage, a PPM enablement signal $enPPM$ generated by the inverter **332** at an output terminal **334** can be a logic zero, i.e., the PPM enablement signal $enPPM=0$. On

the other hand, when the output voltage $V_{sub.out}$ of the comparator **328** is a negative voltage, the PPM enablement signal $enPPM$ can be a logic one, i.e., $enPPM=1$. In the other words, when the electrical potential $V_{sub.ppm}$ at the node **322** is larger than the reference voltage $V_{sub.ref}$ (i.e., $V_{sub.ppm}>V_{sub.ref}$), the PPM enablement signal $enPPM=0$. When the electrical potential $V_{sub.ppm}$ at the node **322** is smaller than the reference voltage $V_{sub.ref}$ (i.e., $V_{sub.ppm}<V_{sub.ref}$), the PPM enablement signal $enPPM=1$.

(39) In some embodiments, the output circuit can also include an optional RC filter **344** connected between the node **322** and the second input terminal **326** of the comparator **328**. The RC filter **344** can be used to filter out unwanted signals within a certain frequency range.

(40) As discussed previously, all the PPM pins of the same PPM group are electrically connected, i.e., all the PPM pins of the same PPM group have the same electrical potential $V_{sub.ppm}$. Therefore, each PPM group only needs one comparator **328** electrically connected at the node **322**. And the PPM enablement signal $enPPM$ indicates the electrical potential $V_{sub.ppm}$ for the PPM group. In the example shown in FIG. 2, the PPM enablement signals for the PPM1 group **210** and the PPM2 group **211** can thereby be presented by the PPM enablement signal $enPPM1$ and the PPM enablement signal $enPPM2$, respectively.

(41) It is noted that the resistance $R_{sub.u}$ and $R_{sub.d}$, the characteristic of the pull-up driver **314** and the pull-down driver **336**, the electrical potential $V_{sub.ppm}$ at node **322** are not necessary to be the same between the PPM1 circuit **202** and the PPM2 circuit **206** shown in FIG. 2. The symbols used for the exemplary PPM circuit **300** are only for illustration purpose and for simplicity to demonstrate the functionality of the PPM circuits.

(42) In some embodiments, the pull-down driver **336** can be replaced by any suitable device that can be implemented as a current source with a pre-determined current level.

(43) During operation, a first control signal **340** can be sent to a gate terminal **316** of the pull-up driver **314** to switch the pull-up driver **314** on or off. For example, if the first control signal **340** has a voltage less than a threshold voltage of the pull-up driver **314**, the pull-up driver **314** can be switched on, and a conductive path can be formed from the power source **312** to the pull-up resistor **318**. The current flowing through the pull-up driver **314** and the pull-up resistor **318** is also referred to as the pull-up current $I_{sub.pull_up}$. If the first control signal **340** has a voltage higher than the threshold voltage of the pull-up driver **314**, the pull-up driver **314** can be switched off.

(44) When a second control signal **342** is sent to a gate terminal **338** of the pull-down driver **336**, the pull-down driver **336** can be switched on or off. For example, if the second control signal **342** has a voltage higher than a threshold voltage of the pull-down driver **336**, the pull-down driver **336** can be switched on, and a conductive path can be formed from the pull-down resistor **320** to the ground. If the second control signal **342** has a voltage less than the threshold voltage of the pull-down driver **336**, the pull-down driver **336** can be switched off.

(45) In some embodiments, the pull-down driver **336** can be operated as a current controller. In this example, when the pull-down driver **336** is switched on, the magnitude of the current flowing through the pull-down resistor **320** (also referred to as pull-down current $I_{sub.pull_dn}$) depends on the second control signal **342**. When the pull-down driver **336** is an nFET, as shown in FIG. 3, the pull-down current $I_{sub.pull_dn}$ can be determined by the voltage level of the second control signal **342** and the trans-conductance of the pull-down driver **336**. The pull-down driver **336** can be switched on or off to signal whether there is a peak power operation performed on the memory die. For example, the voltage level of the second control signal **342** can correspond to a peak current $I_{sub.cc}$ used during the peak power operation. As such, when the memory die is performing the peak power operation, the pull-down driver **336** can be switched on accordingly.

(46) According to another embodiment of the present disclosure, a peak current $I_{sub.cc}$ of a memory die (e.g., the NAND flash memory **100-1**) can correspond to the voltage level of the second control signal **342**, and thereby correspond to the pull-down current $I_{sub.pull_dn}$. Therefore, the pull-down current $I_{sub.pull_dn}$ can function as a current mirror of the peak current

I.sub.cc of the memory die. In some embodiments, the pull-down current I.sub.pull_dn can be proportional to the peak current I.sub.cc. The pull-down current I.sub.pull_dn can be scaled down from the peak current I.sub.cc, a high current level needed by the memory die to perform a peak power operation. For example, if the memory die needs 200 mA of current to perform a peak power operation, the pull-down current I.sub.pull_dn can be 20 μ A. Therefore, the peak power operations can be regulated for each memory die through the pull-down current I.sub.pull_dn. Furthermore, through the die-to-die connections at the PPM contact pads, peak power operations throughout the entire storage system can be coordinated between different memory dies.

(47) During operation, according to some embodiments of the present disclosure, only one pull-up driver **314** is switched on (i.e., enabled) within each PPM group (e.g., within the PPM1 group **210** or the PPM2 group **211** in FIG. 2) and the other pull-up drivers **314** in the same PPM group are switched off. In one example, each of the plurality of memory dies **100-1**, **100-2**, **100-3**, . . . , **100-n** in FIG. 2 can receive a command from the memory controller **20** (as shown in FIG. 1) through a respective communication interface on each memory die. In some embodiments, the command received at the communication interface includes the PPM circuit selecting signal for selecting the PPM1 circuit **202** and/or the PPM2 circuit **206** on the memory die. In response to the PPM circuit selecting signal, the control circuits **70** of the selected memory die can also send control signals (e.g., the first control signal **340** and the second control signal **342**) to the PPM1 circuit **202** and/or the PPM2 circuit **206** on the selected memory die. In this example, the first control signal **340** can be sent by the control circuits **70** to switch on or off the pull-up driver **314** of the PPM1 circuit **202** and/or the PPM2 circuit **206** on the particular memory die. For example, if a memory die is selected according to the PPM circuit selecting signal, the control circuits **70** of the selected memory die can send the first control signal **340** having a voltage less than a threshold voltage of the pull-up driver **314** to switch on the pull-up driver **314**. Otherwise, if a memory die does not receive the PPM circuit selecting signal at the communication interface, the control circuits **70** of the particular memory die can send the first control signal **340** having a voltage higher than a threshold voltage of the pull-up driver **314** to switch off the pull-up driver **314**. As such, current only flows from the power source **312** through one pull-up resistor **318** of the particular memory die in each PPM group. During operation, the pull-down driver **336** can be switched on or off depending on the status of the memory die, and can be independently controlled according to the PPM management scheme discussed below. For example, the NAND flash memory **100-1** (in FIG. 2) can perform a peak power operation using the peak current I.sub.cc when enPPM=0; correspondingly, the pull-down driver **336** of the PPM1 circuit **202-1** is switched on, where a conductive path can be formed from the pull-down resistor **320** through the pull-down driver **336** to the ground. The NAND flash memories **100-1** is prohibited from performing any peak power operation when enPPM=1; correspondingly, the pull-down driver **336** of the PPM1 circuit **202-1** is switched off, where no current can flow through the pull-down driver **336** on the NAND flash memory **100-1**.

(48) The electric potential V.sub.ppm of the node **322** (or the PPM pins) depends on the number of pull-down drivers **336** that are switched on in each PPM group. Because the pull-down driver **336** is switched on when peak power operation can be performed on a memory die, by monitoring the electric potential V.sub.ppm, the number of peak power operations performed in a storage system having multiple memory dies can thereby be regulated.

(49) FIG. 4 shows an equivalent circuit **400** of a PPM group using the PPM circuit **300** in FIG. 3, according to some embodiments of the present disclosure. The PPM group can be the PPM1 group **210** or the PPM2 group **211** in the storage system in FIG. 2. Since only the pull-up driver **314** and the pull-down drivers **336** that are switched on can form conductive paths, FIG. 4 omits those pull-up drivers **314** and pull-down drivers **336** that are switched off. As discussed previously, the node **322** is electrically connected to the PPM pin on each of the memory die in the PPM group, and all the PPM pins in the same PPM group are electrically connected between different memory dies.

The node **322** can be held at the same electrical potential $V_{\text{sub.ppm}}$ in the entire PPM group between different memory dies, and is illustrated as one intersection point to the pull-up resistor **318** in FIG. 4.

(50) In some embodiments, only one pull-up driver **314** is selected to be switched on in one PPM group for peak power management. In some embodiments, the pull-down driver **336** is switched on in the PPM circuit corresponding to the memory die performing the peak power operation, i.e., using the peak current $I_{\text{sub.cc}}$, or using a current greater than a threshold predetermined in the NAND flash memory **100** or the storage system **10**.

(51) In one example, there can be m number of peak power operations in the storage system **10**, where m can be any whole number. For illustration purpose, it is assumed that one memory die can perform one peak power operation. Thus, there can be m number of memory dies (e.g., NAND flash memory **100-1**, **100-2**, . . . , **100- m** in FIG. 2) performing peak power operations simultaneously in the storage system **10**. The pull-down drivers **336** in the same PPM group are connected in parallel with each other. In this configuration, the pull-up current $I_{\text{sub.pull_up}}$ flowing through the pull-up driver **314** that is switched on, is the sum of the pull-down current $I_{\text{sub.pull_dn}}$ (e.g., $I_{\text{sub.pull_dn-1}}$, $I_{\text{sub.pull_dn-2}}$, . . . , $I_{\text{sub.pull_dn- m }$) flowing through each of the pull-down drivers **336** that are switched on. The equivalent circuit **400** functions as a voltage divider, and the electric potential $V_{\text{sub.ppm}}$ of the node **322** can be expressed as a function of m , the number of peak power operation, i.e., the number of memory dies performing peak power operations if one memory die performs one peak power operation:

$$(52) \quad V_{\text{ppm}}(m) = \frac{R_d}{R_d + m \cdot R_u} V_{dd} \quad (1)$$

where $R_{\text{sub.d}}$ and $R_{\text{sub.u}}$ are the resistances of the pull-down resistor **320** and pull-up resistor **318**, respectively, and the power source **312** provides the voltage $V_{\text{sub.dd}}$. In this example, the electric potential $V_{\text{sub.ppm}}$ decreases when m , the number of peak power operations, increases. As the number of peak power operations (i.e., the variable m) changes continuously, the electric potential $V_{\text{sub.ppm}}$ changes continuously accordingly. As such, through an analog circuit like the PPM circuit **300**, information about the peak power operations in each PPM group can be encoded into the electric potential $V_{\text{sub.ppm}}$ of the PPM pins in an analog way.

(53) FIG. 5 illustrates an exemplary peak power management scheme, according to some embodiments of the present disclosure. In this example, the voltage $V_{\text{sub.dd}}$ of the power source **312** is assumed to be 1.2 V. The resistance $R_{\text{sub.d}}$ of the pull-down resistor **320** is assumed to be twice as much as the resistance $R_{\text{sub.u}}$ of the pull-up resistor **318**. It is noted that the assumptions of the voltage $V_{\text{sub.dd}}$ and the resistances $R_{\text{sub.u}}$ and $R_{\text{sub.d}}$ are chosen to provide an illustration of the principles of the peak power management scheme in the present disclosure. A person of ordinary skill in the art can utilize the principle of the peak power management scheme described here with various modifications as suitable for the practical applications. All such modifications and variations are within the scope of the present disclosure.

(54) Referring to FIGS. 4 and 5, at the beginning of peak power management, any one of the pull-up drivers **314** in the respective PPM group can be switched on. If there is no memory die performing peak power operation, none of the pull-down driver **336** is switched on. The electric potential $V_{\text{sub.ppm}}$ of the node **322** is set to the voltage $V_{\text{sub.dd}}$ of the power source **312** through the conductive path established by the specific pull-up driver **314** that is switched on. If $V_{\text{sub.dd}}=1.2\text{V}$, then $V_{\text{sub.ppm}}(0)=1.2\text{ V}$ (at $m=0$).

(55) When one of the memory dies, e.g., the NAND flash memory **100-1** (in FIG. 2), is performing the peak power operation, the memory die consumes the peak current $I_{\text{sub.cc-1}}$. As discussed previously, the second control signal **342-1** can be generated to switch on the pull-down driver **336-1** to enable the peak power operation on the NAND flash memory **100-1**. As such, the pull-down current $I_{\text{sub.pull_dn-1}}$ can flow through the pull-down driver **336-1** and the pull-down resistor **320-1**. The electric potential $V_{\text{sub.ppm}}$ at the node **322** is changed to

$$(56) V_{ppm}(1) = \frac{R_d}{R_d + R_u} V_{dd} (atm = 1).$$

Assuming the resistance $R_{sub.d}$ is twice as much as the resistance $R_{sub.u}$, the electric potential $V_{sub.ppm}(1) = \frac{2}{3} V_{sub.dd} = 0.8$ V.

(57) When another memory die, e.g., the NAND flash memory **100-2** (in FIG. 2), is also performing the peak power operation, there are two memory dies ($m=2$) in the storage system **10** are now using the peak current $I_{sub.cc}$, i.e., the peak current $I_{sub.cc-1}$ on the NAND flash memory **100-1** and the peak current $I_{sub.cc-2}$ on the NAND flash memory **100-2**. In some embodiments, the peak current $I_{sub.cc-1}$ of the NAND flash memory **100-1** and the peak current $I_{sub.cc-2}$ of the NAND flash memory **100-2** can be the same. In some embodiments, the peak current $I_{sub.cc-1}$ of the NAND flash memory **100-1** and the peak current $I_{sub.cc-2}$ of the NAND flash memory **100-2** can be the different. As discussed previously, the second control signal **342-2** can be generated to switch on the pull-down driver **336-2** to enable the peak power operation on the NAND flash memory **100-2**. As such, the pull-down current $I_{sub.pull_dn-2}$ can flow through the pull-down driver **336-2** and the pull-down resistor **320-2**. The pull-down current $I_{sub.pull_dn-1}$ and the pull-down current $I_{sub.pull_dn-2}$ flow through the pull-down resistor **320-1** and the pull-down resistor **320-2**, respectively, where the pull-down resistor **320-1** and the pull-down resistor **320-2** are connected in parallel. The electric potential $V_{sub.ppm}$ at the node **322** is then changed to

$$(58) V_{ppm}(2) = \frac{R_d}{R_d + 2 \cdot \text{Math. } R_u} V_{dd} (atm = 2).$$

If assuming that the resistance $R_{sub.d}$ is twice as much as the resistance $R_{sub.u}$, the electric potential $V_{sub.ppm}(2) = \frac{1}{2} V_{sub.dd} = 0.6$ V.

(59) When a third memory die, e.g., the NAND flash memory **100-3** (in FIG. 2), is also performing the peak power operation, i.e., three memory dies ($m=3$) in the storage system **10** are consuming the peak current $I_{sub.cc}$, the electric potential $V_{sub.ppm}$ at the node **322** is then changed to

$$(60) V_{ppm}(3) = \frac{R_d}{R_d + 3 \cdot \text{Math. } R_u} V_{dd} (atm = 3).$$

If assuming that the resistance $R_{sub.d}$ is twice as much as the resistance $R_{sub.u}$, the electric potential $V_{sub.ppm}(3) = \frac{2}{5} V_{sub.dd} = 0.48$ V.

(61) Using the Equation (1), the electric potential $V_{sub.ppm}$ can be calculated if four memory dies ($m=4$) in the storage system **10** are using the peak current $I_{sub.cc}$, i.e. performing the peak power operations. The electric potential

$$(62) V_{ppm}(4) = \frac{R_d}{R_d + 4 \cdot \text{Math. } R_u} V_{dd};$$

or $V_{sub.ppm}(4) = \frac{1}{3} V_{sub.dd} = 0.4$ V (if $V_{sub.dd} = 1.2$ V, $R_{sub.d} = 2R_{sub.u}$). The electric potentials $V_{sub.ppm}(m)$ are listed in FIG. 5 for various number of peak power operations (i.e., 0, 1, 2, 3, 4, . . . , $m-1$, m).

(63) As discussed above, the electric potential $V_{sub.ppm}$ of a PPM group depends on the number m of peak power operations. According to Equation (1) and FIG. 5, the electric potential $V_{sub.ppm}$ decreases when the number of the peak power operations increases. Therefore, the reference voltage $V_{sub.ref}$ for the comparator **328** (in FIG. 3) can be chosen such that the PPM enablement signal $enPPM$ can be set at logic zero, i.e., $enPPM=0$ when the electric potential $V_{sub.ppm}$ is higher than the reference voltage $V_{sub.ref}$, i.e., when the number of memory dies performing peak power operations is less than a predetermined number. And the PPM enablement signal $enPPM$ can be set at logic one, i.e., $enPPM=1$ when the electric potential $V_{sub.ppm}$ is not higher than the reference voltage $V_{sub.ref}$, i.e., when the number of memory dies performing peak power operations has reached (equal or greater than) the predetermined number. In the other words, the reference voltage $V_{sub.ref}$ can be programmed to correspond to the maximum number of peak power operations allowed in a storage system. Accordingly, when the PPM enablement signal $enPPM=0$, the storage system can still provide additional peak current $I_{sub.cc}$ to at least one of the memory dies, i.e., having enough power (or current) to provide at least one additional memory die to perform peak power operation. On the contrary, when the PPM enablement signal $enPPM=1$, the maximum number of peak power operations have been reached for the storage system. The storage

system has reached its power (or current) limit and cannot provide additional peak current $I_{sub.cc}$ to any of the memory dies to perform any additional peak power operation.

(64) For a storage system where each memory die includes more than one PPM group, one reference voltage $V_{sub.ref}$ can be chosen for each PPM group to output a PPM enablement signal $enPPM$ for controlling a maximum number of peak power operations. In the example of the storage system in FIG. 2, the PPM1 group can have a reference voltage $V_{sub.ref-1}$ programed to correspond to a number $m_{sub.1}$ —the maximum number of peak power operations allowed in the PPM1 group. Similarly, the PPM2 group can have a reference voltage $V_{sub.ref-2}$ programed to correspond to a number $m_{sub.2}$ —the maximum number of peak power operations allowed in the PPM2 group.

(65) For illustration purpose below, it is assumed that the maximum number of peak power operations in a storage system (or in each memory group) is the number m , and each memory die can perform a peak power operation at a time. It is noted that these assumptions are chosen for simplicity and should not limit the scope of this disclosure. For example, each memory die can perform two or more peak power operations simultaneously.

(66) According to Equation (1), if there are $m-1$ number of peak power operations, the electric potential $V_{sub.ppm}$ can be expressed as:

$$(67) \quad V_{ppm}(m-1) = \frac{R_d}{R_d + (m-1) \cdot \text{Math. } R_u} V_{dd} \quad (2)$$

Thus, the reference voltage $V_{sub.ref}(m)$, corresponding to the maximum number m of peak power operations, can be set to a voltage in a range between the electric potential $V_{sub.ppm}(m)$ and the electric potential $V_{sub.ppm}(m-1)$:

$$(68) \quad \frac{R_d}{R_d + m \cdot \text{Math. } R_u} V_{dd} < V_{ref}(m) < \frac{R_d}{R_d + (m-1) \cdot \text{Math. } R_u} V_{dd} \quad (3)$$

(69) In the example shown in FIG. 5, if the maximum number of peak power operations allowed is 2, the reference voltage $V_{sub.ref}$ can be a voltage in a range between $V_{sub.ppm}(2)=0.6$ V and $V_{sub.ppm}(1)=0.8$ V, where $V_{sub.ref}(2)=0.7$ V is chosen in FIG. 5 as an example. If the electric potential $V_{sub.ppm}$ is higher than the reference voltage $V_{sub.ref}=0.7$ V, the PPM enablement signal $enPPM=0$ is generated at the output terminal 334 (see FIG. 3), indicating that the number of peak power operations is 0 or 1, less than the maximum number of 2. The storage system can have at least one additional peak power operation performed by at least one of the memory dies. When there are two peak power operations performed simultaneously, i.e., two memory dies are consuming the peak current $I_{sub.cc}$, the electric potential $V_{sub.ppm}=0.6$ V, lower than the reference voltage $V_{sub.ref}=0.7$ V. The PPM enablement signal $enPPM=1$ is generated accordingly, indicating that the maximum number of peak power operations of 2 has been reached. The NAND storage system/the PPM group can then prohibit or disable additional memory die to perform a peak power operation. As such, the PPM enablement signal $enPPM$ can function as an indicator for the number of peak power operations performed in the storage system with multiple memory dies. Through analog PPM circuit 300 and voltages (e.g., analog voltages) associated with the electric potential $V_{sub.ppm}$, the reference voltage $V_{sub.ref}$, the second control signal 342, etc., multiple peak power operations can be controlled simultaneously.

(70) As discussed above, the PPM enablement signal $enPPM$ for each PPM group can be generated by comparing the reference voltage $V_{sub.ref}$ with the electric potential $V_{sub.ppm}$ of the PPM pins (or at the node 322) that are electrically connected between the memory dies. The PPM enablement signal $enPPM$ is set to 0 if the electric potential $V_{sub.ppm}$ of the PPM pins is higher than the reference voltage $V_{sub.ref}$. The PPM enablement signal $enPPM$ is set to 1 if the electric potential $V_{sub.ppm}$ of the PPM pins is not higher than the reference voltage $V_{sub.ref}$. Because the electric potential $V_{sub.ppm}$ depends on the status of the pull-down drivers in the PPM group, and the pull-down driver is switched on when peak power operation is performed on the corresponding memory die, thus the reference voltage can be selected based on a maximum m number of peak power operations for the storage system.

(71) FIG. 6 illustrates a peak power check routine **600** associated with the peak power management system **200** in FIG. 2 and the PPM circuit **300** in FIG. 3, according to some embodiments of the present disclosure. The PPM schemes shown in FIGS. 4 and 5 are used to determine the reference voltage $V_{sub.ref}$ and generate the PPM enablement signal $enPPM$ to indicate the maximum number of peak power operations allowed on a storage system. It should be understood that the peak power check (PPC) routine **600** are not exhaustive and that other operation steps can be performed as well before, after, or between any of the illustrated operation steps. In some embodiments, some operation steps of the PPC routine **600** can be omitted or other operation steps can be included, which are not described here for simplicity. In some embodiments, operation steps of the PPC routine **600** can be performed in a different order and/or vary.

(72) The PPC routine **600** provides an exemplary method of managing peak power usage for a storage system with one or more memory dies, where each memory die includes at least one PPM circuit. The example below is shown for a storage system, e.g., in FIG. 2, where each memory die includes two PPM circuits, the PPM1 circuit **202** and the PPM2 circuit **206**. Accordingly, the storage system includes two PPM groups, the PPM1 group **210** and the PPM2 group **211** for checking and regulating peak power operations performed by the memory dies. However, the method can be extended to a storage system where each memory die includes one or more PPM circuits.

(73) The PPC routine **600** can be implemented before a memory die starts to perform a peak power operation such that the total power (or current) consumed by a storage system can be regulated and controlled to below a predetermined value.

(74) The PPC routine **600** starts at operation step **S605**, when a storage system (e.g., the NAND storage system **10** in FIG. 1) determines that a memory die (e.g., the NAND flash memory **100-1** in FIG. 2) is ready to start a peak power operation. At the operation step **S605**, the PPM circuits (e.g., the first PPM circuit **202-1** and the second PPM circuit **206-1**) on the NAND flash memory **100-1** are at a reset state. At the reset state, the pull-down drivers **336** in the PPM1 circuit **202-1** and the PPM2 circuit **206-1** are switched off. At the operation step **S605**, one of the pull-up drivers **314** in the PPM1 group **210** can be switched on if none of the pull-up drivers **314** in the PPM1 group **210** has been switched on. Similarly, one of the pull-up drivers **314** in the PPM2 group **211** can be switched on if none of the pull-up drivers **314** in the PPM2 group **211** has been switched on.

(75) At operation step **S610**, the PPC routine **600** enters a first check point, where the PPM enablement signal $enPPM1$ of the PPM1 group is verified. If the PPM enablement signal $enPPM1$ is not zero (e.g., $enPPM1=1$), the PPC routine **600** pauses at operation step **S615** and the PPM enablement signal $enPPM1$ is polled and verified again at operation step **S610**. As discussed previously, when the PPM enablement signal $enPPM1=1$, a maximum number of peak power operations have been reached, and the storage system cannot support any additional peak power operations or provide any additional operation using the peak current $I_{sub.cc}$. Thus, when the PPM enablement signal $enPPM1=1$, the NAND flash memory **100-1** is prohibited to perform a peak power operation and must wait until the status of the $enPPM1$ is changed.

(76) When the PPM enablement signal $enPPM1$ is zero, it is indicated that the storage system can support at least one additional peak power operation. The PPC routine **600** can then perform operation step **S620**.

(77) At operation step **S620**, the pull-down driver **336-1** of the PPM1 circuit **202-1** on the NAND flash memory **100-1** can be switched on. If the resulting pull-down current $I_{sub.pull_dn-1}$ flowing through the pull-down driver **336-1** of the PPM1 circuit **202-1** can reduce the electric potential $V_{sub.ppm}$ of the PPM1 group to equal or below the reference voltage $V_{sub.ref}$, the PPM enablement signal $enPPM1$ is set to 1 (i.e., $enPPM1=1$). As a result, no more other memory dies requesting peak power operation and running the PPC routine **600** can pass the first check point at the operation step **S610** after the PPM enablement signal $enPPM1$ is set to 1. If the resulting pull-down current $I_{sub.pull_dn-1}$ flowing through the pull-down driver **336-1** of the PPM1 circuit **202-**

1 does not lower the electric potential $V_{sub.ppm}$ of the PPM1 group to a value equal or less than the reference voltage $V_{sub.ref}$, the PPM enablement signal $enPPM1$ remains as zero (i.e., $enPPM1=0$). Other memory dies requesting peak power operation and running the PPC routine 600 can still pass the first check point at the operation step S610 until the PPM enablement signal $enPPM1$ is set to 1. As such one or more memory dies can run the PPC routine 600 simultaneously, and the peak power operations can be performed and coordinated more efficiently.

(78) Next, at operation step S625, the PPC routine 600 is paused for a waiting period of a first delay period $t_{sub.dl_1}$, wherein the first delay period $t_{sub.dl_1}$ is different for each memory die. In the other words, the NAND flash memory 100-1 is assigned with the first delay period $t_{sub.dl_1}$ having a value different from any other NAND flash memories 100-2, 100-3, . . . 100- n in the same storage system. As such, when more than one memory die runs the PPC routine 600 simultaneously, each memory die waits for a different time period at the operation step S625. As discussed previously, the PPM pins of the same PPM group are electrically connected with each other and thus the memory dies in the same PPM group are controlled by the same electric potential $V_{sub.ppm}$. When the PPM enablement signal $enPPM1$ is zero (i.e., $enPPM1=0$) at operation step S610, one or more memory dies of the same storage system that is running the PPC routine 600 can enable (or switch on) the pull-down driver 336 at the same time. By imposing a unique time delay for each memory die, memory dies can enter the subsequent operation step S630 one at a time. The memory dies completing the operation step S620 at the same time, can be desynchronized at the operation step S625. The operation step S625 can ensure that only one additional peak power operation can be allowed at any given time.

(79) When the NAND flash memory 100-1 finishes the waiting period of the first delay period $t_{sub.dl_1}$, a second check point at operation step S630 can be resumed.

(80) At operation step S630, the PPM enablement signal $enPPM2$ of the PPM2 group 211 is verified. If the PPM enablement signal $enPPM2$ is not zero (e.g., when $enPPM2=1$), the PPC routine 600 pauses at operation step S635 and the PPM enablement signal $enPPM2$ is polled and verified again at operation step S640. As discussed previously, when the PPM enablement signal $enPPM2=1$, a maximum number of peak power operations have been reached, and the storage system cannot support any additional peak power operation. Thus, when the PPM enablement signal $enPPM2=1$, the NAND flash memory 100-1 is prohibited to perform a peak power operation and must wait until the status of the $enPPM2$ is changed.

(81) In some embodiments, the PPM enablement signal $enPPM1$ of the PPM1 group is zero at operation step S610, but the PPM enablement signal $enPPM2$ of the PPM2 group is not zero at operation step S630. In this example, additional peak power operation is started between the time period of the operation step S610 and operation step S630 by another memory die (e.g., NAND flash memory 100-2) in the storage system. For example, the NAND flash memory 100-2 completes operation step S625 before the NAND flash memory 100-1. By implementing operation steps S635 and S640, the memory die that has passed the first check point at operation step S610 can wait for its turn to perform a peak power operation.

(82) When the PPM enablement signal $enPPM2$ is zero (i.e., $enPPM2=0$) at operation step S640, operation step S625 can be performed again to implement the first delay period $t_{sub.dl_1}$. And operation step S630 is performed again to verify if the PPM enablement signal $enPPM2=0$.

(83) When the PPM enablement signal $enPPM2=0$ at operation step S630, the PPC routine 600 starts operation step S645, where the pull-down driver 336 of the PPM2 circuit 206-1 is switched on for the NAND flash memory 100-1.

(84) At operation step S650, the NAND flash memory 100-1 starts to perform the peak power operation with the peak current $I_{sub.cc}$. If the maximum number of peak power operations have been reached after the NAND flash memory 100-1 starts the peak power operation, the electric potential $V_{sub.ppm}$ of the PPM2 group can be less than the reference voltage $V_{sub.ref}$ and thereby the PPM enablement signal $enPPM2$ is set to 1 (i.e., $enPPM2=1$). If the maximum number of peak

power operations have not been reached after the NAND flash memory **100-1** starts the peak power operation, the electric potential $V_{sub.ppm}$ of the PPM2 group is still higher than the reference voltage $V_{sub.ref}$. The PPM enablement signal $enPPM2$ remains as zero (i.e., $enPPM2=0$).

(85) At operation step **S655**, after completing the peak power operation, the pull-down driver **336** of the PPM2 circuit **206-1** can be disabled (i.e., switched off) for the NAND flash memory **100-1**. Accordingly, the PPM enablement signal $enPPM2$ for the PPM2 group can be reset to zero again.

(86) At operation step **S660**, the PPC routine **600** performs a second delay period $t_{sub.dl_2}$, where the second delay period $t_{sub.dl_2}$ can be a predetermined time period, for example, $2\ \mu s$. The second delay period $t_{sub.dl_2}$ is chosen such that the memory dies, which have completed operation steps **S610** and **S620** and are waiting at the operation steps **S635** and **S640**, can complete the peak power operation at operation step **S650** before the PPC routine **600** allows other memory dies to pass through the first check point at the operation step **S610**. As such, memory dies that are delayed at the operation step **S625** can be queued for peak power operations and complete operation steps **S645-S655** one at a time, prior to any memory die that have not entered the first check point at operation step **S610**.

(87) At operation step **S665**, the pull-down driver **336** of the PPM1 circuit **202-1** on the NAND flash memory **100-1** is disabled (i.e., switched off). Accordingly, the PPM enablement signal $enPPM1$ for the PPM1 group can be reset to zero again. Another batch of multiple memory dies can subsequently pass the first check point at operation step **S610** and are queued for peak power operations.

(88) Through an analog processing of the electric potential $V_{sub.ppm}$ of the PPM pins connected between multiple memory dies, analog peak power management can be implemented. The PPM enablement signal $enPPM$ can be used to encode with information regarding the number of peak power operations performed in a storage system. Multiple peak power operations below a maximum number can be performed simultaneously in a storage system with multiple memory dies. By using the PPC routine with two PPM groups, multiple memory dies can be queued and staggered for peak power operations once the power/current budget is available for the storage system.

(89) FIG. 7 illustrates an exemplary PPM circuit **700**, according to some embodiments of the present disclosure, where each of the PPM1 circuits **202** and each of the PPM2 circuits **206** shown in FIG. 2 can be similar to the PPM circuit **700**. The PPM circuit **700** is similar to the PPM circuit **300** shown in FIG. 3, except that the pull-up resistor **318** in FIG. 3 can be replaced by a pull-up current source **714** and that the inverter **332** of the output circuit in FIG. 3 can be removed. The pull-up current source **714** can be any current source that provide constant current. In some embodiments, the pull-up current source **714** can be a transistor, for example, a field-effect transistor (FET). In FIG. 7, a p-channel MOSFET is shown as an example. However, the pull-up current source **714** is not so limiting. For example, the pull-up current source **714** can be a bipolar transistor, an n-channel MOSFET, etc. The functionality of the other components of the PPM circuit **700** is similar to the PPM circuit **300** described with respect to FIG. 3, and only the differences will be discussed below.

(90) In this example, the magnitude of the pull-up current $I_{sub.pull_up}$ flowing through the pull-up current source **714** can be controlled through the pull-up current source **714**. In one example, the pull-up current $I_{sub.pull_up}$ can be determined by a voltage level at a gate terminal **716** and the trans-conductance of the pull-up current source **714**. In this example, the pull-up current $I_{sub.pull_up}$ can signal the peak power operation performed on the memory die. For example, the pull-up current $I_{sub.pull_up}$ can be proportional to the peak current $I_{sub.cc}$. The pull-up current $I_{sub.pull_up}$ can be scaled down from the peak current $I_{sub.cc}$. Therefore, the peak power operations can be regulated for each memory die through the pull-up current $I_{sub.pull_up}$. Similar to the PPM circuit **300** in FIG. 3, through the die-to-die connections at the PPM contact pads, peak power operations throughout the entire storage system can be coordinated between different

memory dies using the PPM circuit **700**.

(91) In this example, one pull-down driver **336** can be switched on (i.e., enabled) within each PPM group (e.g., within the PPM1 group **210** or the PPM2 group **211** in FIG. 2) during operation and the other pull-down drivers **336** in the same PPM group can be switched off. As such, in each PPM group, current only flows from the pull-down resistor **320** through the pull-down driver **336**, which is switched on, to the ground. In this example, in response to the PPM circuit selecting signal received at the communication interface, the control circuits **70** of the memory die can send the second control signal **342** to switch on or off the pull-down driver **336** of the PPM1 circuit **202** and/or the PPM2 circuit **206** on the memory die. For example, if a particular memory die is selected according to the PPM circuit selecting signal, the control circuits **70** of the particular memory die can send the second control signal **342** having a voltage higher than a threshold voltage of the pull-down driver **336** to switch on the pull-down driver **336**. Otherwise, if a memory die does not receive the PPM circuit selecting signal at the communication interface, the control circuits **70** of the memory die can send the second control signal **342** having a voltage lower than a threshold voltage of the pull-down driver **336** to switch off the pull-down driver **336**. During operation, the pull-up driver **314** can be switched on or off depending on the status of the memory die, and can be independently controlled according to the PPM management scheme discussed below. For example, when the NAND flash memory **100-1** (in FIG. 2) performs a peak power operation using the peak current $I_{sub.cc}$, the pull-up driver **314** of the PPM1 circuit **202-1** can be switched on. A conductive path can be formed from the power source **312** through the pull-up driver **314** to the pull-up current source **714**, where the pull-up current $I_{sub.pull-up}$ can be a current mirror of the peak current $I_{sub.cc}$. The NAND flash memories **100-1** is prohibited from performing any peak power operation when $enPPM=1$; correspondingly, the pull-up driver **314** of the PPM1 circuit **202-1** is switched off, where no current can flow through the pull-up driver **314** on the NAND flash memory **100-1**.

(92) The electric potential $V_{sub.ppm}$ of the node **322** (or the PPM pins) depends on the number of pull-up drivers **314** that are switched on in each PPM group. Because a peak power operation can be performed on a memory die when the pull-up driver **314** is switched on, by monitoring the electric potential $V_{sub.ppm}$, the number of peak power operations performed in a storage system having multiple memory dies can thereby be regulated.

(93) FIG. 8 shows an equivalent circuit **800** of a PPM group using PPM circuit **700** shown in FIG. 7, according to some embodiments of the present disclosure. In operation, since only the pull-up drivers **314** and the pull-down driver **336** that are switched on can form conductive paths, FIG. 8 omits those pull-up drivers **314** and pull-down drivers **336** that are switched off. As discussed previously, the node **322** is electrically connected to the PPM pin on the memory die, and all the PPM pins in the same PPM group are electrically connected between different memory dies. The node **322** can be held at the same electrical potential $V_{sub.ppm}$ in the entire PPM group between different memory dies, and is illustrated as one intersection point to the pull-down resistor **320** in FIG. 8.

(94) While only one pull-down driver **336** is switched on in one PPM group for peak power management, the pull-up drivers **314** (e.g., **314-1**, **314-2**, . . . , **314-m**) can be switched on according to the memory die performing the peak power operation, i.e., using the peak current $I_{sub.cc}$. Assuming that one memory die can perform one peak power operation, then m number of memory dies (e.g., NAND flash memory **100-1**, **100-2**, . . . , **100-m** in FIG. 2) can perform m number of peak power operations simultaneously in the storage system **10**. In the configuration in FIG. 8, the pull-up driver **314** is connected in series with the pull-up current source **714** on one memory die. In the same PPM group, the pull-up drivers **314** (e.g., **314-1**, **314-2**, . . . , **314-m**) are connected in parallel, and the pull-up current sources **714** (e.g., **714-1**, **714-2**, . . . , **714-m**) are connected in parallel. In this configuration, the pull-down current $I_{sub.pull_dn}$ flowing through the pull-down driver **336** is the sum of the pull-up currents $I_{sub.pull-up}$ (e.g., $I_{sub.pull_up-1}$, $I_{sub.pull_up-2}$, . . .

, I.sub.pull_up-m) flowing through each of the pull-up current sources **714** and the pull-up drivers **314** that are switched on. The equivalent circuit **800** functions as a voltage divider, and the electric potential V.sub.ppm of the node **322** can be expressed as a function of m, the number of peak power operation, i.e., the number of memory dies performing peak power operations if one memory die performs one peak power operation:

$$V_{\text{sub.ppm}}(m) = m \cdot i \cdot R_{\text{sub.d}} \quad (4)$$

where R.sub.d is the resistances of the pull-down resistor **320**, and i is the magnitude of the pull-up current I.sub.pull-up, which can be controlled through the pull-up current source **714** (e.g., **714-1**, **714-2**, . . . , **714-m**). In some embodiments, the magnitude i of the pull-up current I.sub.pull-up (e.g., I.sub.pull_up-1, I.sub.pull_up-2, . . . , I.sub.pull_up-m) can be different on different memory dies and a person of skill in the art can modify the Equation (4) accordingly. As shown in Equation (4), by using the pull-up current source **714** in FIG. 7 instead of the pull-up resistor **318** in FIG. 3, the electrical potential V.sub.ppm at the node **322** (or at the first and second PPM contact pads **204/208**) can be proportional to the number m of the peak power operations. In this example, the electrical potential V.sub.ppm can have a liner relationship with the number m of the peak power operations by using the pull-up current source **714**. In this example, the electric potential V.sub.ppm increases when m, the number of peak power operations, increases. As the number of peak power operations (i.e., the variable m) changes continuously, the electric potential V.sub.ppm changes continuously accordingly. Similar to embodiment shown in FIG. 3, through an analog circuit like the PPM circuit **700**, information about the peak power operations in each PPM group can be encoded into the electric potential V.sub.ppm of the PPM pins in an analog way in embodiment shown in FIG. 7.

(95) FIG. 9 illustrates an exemplary peak power management scheme using the PPM circuit **700** and its equivalent circuit **800**, according to some embodiments of the present disclosure. In this example, the resistance R.sub.d of the pull-down resistor **320** is assumed to be 1000 ohm and the magnitude i of the pull-up current I.sub.pull-up is assumed to be 160 μ A. It is noted that these assumptions are chosen to provide an illustration of the principles of the peak power management scheme in the present disclosure. A person of ordinary skill in the art can utilize the principle of the peak power management scheme described here with various modifications as suitable for the practical applications. All such modifications and variations are within the scope of the present disclosure.

(96) Referring to FIGS. 8 and 9, to start peak power management, any one of the pull-down driver **336** in the respective PPM group can be switched on. If there is no memory die performing peak power operation, none of the pull-up drivers **314** is switched on in the PPM group. The electric potential V.sub.ppm of the node **322** is connected to the ground through the conductive path established by the pull-down resistor **320** and the specific pull-down driver **336** that is switched on. As such, $V_{\text{sub.ppm}}(0) = 0$ V (at m=0).

(97) When one of the memory dies, e.g., the NAND flash memory **100-1** (in FIG. 2), is performing the peak power operation, the memory die consumes the peak current I.sub.cc-1. The first control signal **340-1** can be generated to switch on the pull-up driver **314-1** of the PPM circuit on the NAND flash memory **100-1** to signal the peak power operation on the NAND flash memory **100-1**. The pull-up current I.sub.pull_up-1 can flow from the power source **312** through the pull-up driver **314-1**, the pull-up current source **714-1**, the pull-down resistor **320** and the pull-down driver **336** to the ground. The electric potential V.sub.ppm at the node **322** is changed to $V_{\text{sub.ppm}}(1) = i \cdot R_{\text{sub.d}} = 0.16$ V (at m=1).

(98) When another memory die, e.g., the NAND flash memory **100-2** (in FIG. 2), is also performing the peak power operation, the first control signal **340-2** can be generated to switch on the pull-up driver **314-2** to signal the peak power operation on the NAND flash memory **100-2**. As such, the pull-up current I.sub.pull_up-2 can flow through the pull-up driver **314-2** and the pull-up current source **714-2**. The pull-up current I.sub.pull_up-1 and the pull-up current I.sub.pull_up-2

flow through the pull-down resistor **320** and the pull-down driver **336** as the pull-down current $I_{sub_pull_dn}$. The electric potential V_{sub_ppm} at the node **322** is then changed to $V_{sub_ppm}(2)=2*i*R_{sub_d}=0.32$ V (at $m=2$).

(99) Similarly, when a third memory die, e.g., the NAND flash memory **100-3** (in FIG. 2), is also performing the peak power operation, the electric potential V_{sub_ppm} at the node **322** is then changed to $V_{sub_ppm}(3)=3*i*R_{sub_d}=0.48$ V (at $m=3$). Using the Equation (4), the electric potential V_{sub_ppm} can be calculated if four memory dies ($m=4$) in the storage system **10** are using the peak current I_{sub_cc} , i.e. performing the peak power operations. The electric potential $V_{sub_ppm}(4)=4*i*R_{sub_d}=0.64$ V. The electric potentials $V_{sub_ppm}(m)$ are listed in FIG. 9 for various number of peak power operations (i.e., 0, 1, 2, 3, 4, . . . , $m-1$, m), where m is a positive integer.

(100) As such, the electric potential V_{sub_ppm} of the node **322** or the PPM contact pads depends on the number m of peak power operations. According to Equation (4) and FIG. 9, the electric potential V_{sub_ppm} increases when the number m of the peak power operations increases. Therefore, the reference voltage V_{sub_ref} for the comparator **328** (in FIG. 7) can be chosen such that the PPM enablement signal $enPPM$ at the output terminal **330** of the comparator **328** can be set at, for example, logic zero ($enPPM=0$) when the electric potential V_{sub_ppm} is lower than the reference voltage V_{sub_ref} , i.e., when the number of memory dies performing peak power operations is less than a predetermined number. And the PPM enablement signal $enPPM$ can be set at $enPPM=1$ when the electric potential V_{sub_ppm} is not lower than the reference voltage V_{sub_ref} , i.e., when the number of memory dies performing peak power operations has reached (equal or greater than) the predetermined number.

(101) Similar to the embodiment shown in FIG. 3, the reference voltage $V_{sub_ref}(m)$, corresponding to the maximum number m of peak power operations, can be set to a voltage in a range between the electric potential $V_{sub_ppm}(m)$ and the electric potential $V_{sub_ppm}(m-1)$:
$$(m-1)*i*R_{sub_d} < V_{sub_ref}(m) < m*i*R_{sub_d} \quad (5)$$

(102) In the example shown in FIG. 9, if the maximum number of peak power operations allowed is 2, the reference voltage V_{sub_ref} can be a voltage in a range between $V_{sub_ppm}(2)=0.32$ V and $V_{sub_ppm}(1)=0.16$ V, where $V_{sub_ref}(2)=0.24$ V can be chosen in FIG. 9 as an example. If the electric potential V_{sub_ppm} is less than the reference voltage $V_{sub_ref}=0.24$ V, the PPM enablement signal $enPPM=0$ is generated at the output terminal **330** (see FIG. 7), indicating that the number of peak power operations is 0 or 1, less than the maximum number of 2. The storage system can have at least one additional peak power operation performed by at least one of the memory dies. When there are two peak power operations performed simultaneously, e.g., two memory dies are consuming the peak current I_{sub_cc} , the electric potential $V_{sub_ppm}=0.32$ V, higher than the reference voltage $V_{sub_ref}=0.24$ V. The PPM enablement signal $enPPM=1$ can be generated accordingly, indicating that the maximum number of peak power operations of 2 has been reached. The NAND storage system can then prohibit additional memory die to perform a peak power operation. As such, the PPM enablement signal $enPPM$ can function as an indicator for the number of peak power operations performed in the storage system with multiple memory dies. Through analog PPM circuit **700** and voltages (e.g., analog voltages) associated with the electric potential V_{sub_ppm} , the reference voltage V_{sub_ref} , gate voltage applied to the pull-up current source **714**, etc., multiple peak power operations can be controlled simultaneously.

(103) FIG. 10 illustrates a peak power check routine **1000** associated with the peak power management system **200** in FIG. 2 and the PPM circuit **700** in FIG. 7, according to some embodiments of the present disclosure. The PPM schemes shown in FIGS. 8 and 9 are used to determine the reference voltage V_{sub_ref} and generate the PPM enablement signal $enPPM$ to indicate the maximum number of peak power operations allowed on a storage system. It should be understood that the peak power check (PPC) routine **1000** are not exhaustive and that other operation steps can be performed as well before, after, or between any of the illustrated operation

steps. In some embodiments, some operation steps of the PPC routine **1000** can be omitted or other operation steps can be included, which are not described here for simplicity. In some embodiments, operation steps of the PPC routine **1000** can be performed in a different order and/or vary.

(104) The PPC routine **1000** is similar to the PPC routine **600** shown in FIG. 6, except that the PPM circuit **300** in FIG. 3 is replaced by PPM circuit **700** in FIG. 7. While the electric potential $V_{sub.ppm}$ decreases with the number m of peak power operations using PPM circuit **300** (see Equation (2) and FIG. 5), the electric potential $V_{sub.ppm}$ proportionally increases with the number m of peak power operations using PPM circuit **700** (see Equation (4) and FIG. 9). Additionally, by using the pull-up current source **714** instead of the pull-up resistor **318**, the electric potential $V_{sub.ppm}$ is in linear relationship with the number m of peak power operations (see Equation (4)). The description below for the PPC routine **1000** will be focused on the difference from the PPC routine **600**. Similar operations will be omitted for simplicity.

(105) At the operation step **S1005**, the PPM circuits **700** of the PPM1 group **210** and the PPM2 group **211** (in FIG. 2) can be reset first. At the reset state, the pull-up drivers **314** are all switched off. One of the pull-down drivers **336** in the PPM1 group **210** can be switched on. Similarly, one of the pull-down drivers **336** in the PPM2 group **211** can be switched on.

(106) At operation step **S1010**, the PPC routine **1000** enters a first check point, where the PPM enablement signal $enPPM1$ of the PPM1 group is verified. Similar to the operation step **S610** in FIG. 6, if the PPM enablement signal $enPPM1$ is not zero (e.g., $enPPM1=1$), the PPC routine **1000** will be paused because a maximum number of peak power operations have been reached, and the storage system cannot support any additional peak power operations or provide any additional operation using the peak current $I_{sub.cc}$. The PPM enablement signal $enPPM1$ is polled at operation step **S1015**, and verified again at operation step **S1010**. As such, when the PPM enablement signal $enPPM1=1$, the NAND flash memory **100-1** (in FIG. 2) is prohibited to perform a peak power operation and must wait until the status of the $enPPM1$ is changed.

(107) When the PPM enablement signal $enPPM1$ is zero, it is indicated that the storage system can support at least one additional peak power operation. The PPC routine **1000** moves to the operation step **S1020**.

(108) At operation step **S1020**, the pull-up driver **314-1** of the PPM1 circuit **202-1** on the NAND flash memory **100-1** can be switched on. If the resulting pull-up current $I_{sub.pull_up-1}$ that flows through the pull-up driver **314-1** and the pull-up current source **714-1** of the PPM1 circuit **202-1** increases the electric potential $V_{sub.ppm}$ of the PPM1 group to equal or above the reference voltage $V_{sub.ref}$, the PPM enablement signal $enPPM1$ will be set to 1 (i.e., $enPPM1=1$). As a result, no more other memory dies requesting peak power operation and running the PPC routine **1000** can pass the first check point at the operation step **S1010** after the PPM enablement signal $enPPM1$ is set to 1. If the resulting pull-up current $I_{sub.pull_up-1}$ does not increase the electric potential $V_{sub.ppm}$ of the PPM1 group to equal or above the reference voltage $V_{sub.ref}$, the PPM enablement signal $enPPM1$ remains as zero (i.e., $enPPM1=0$). Other memory dies requesting peak power operation and running the PPC routine **1000** can still pass the first check point at the operation step **S1010** until the PPM enablement signal $enPPM1$ is set to 1. As such one or more memory dies can run the PPC routine **1000** simultaneously, and the peak power operations can be performed and coordinated more efficiently.

(109) Similar to the operation step **S625** of the PPC routine **600** (FIG. 6), the PPC routine **1000** (FIG. 10) pauses at the operation step **S1025** for the waiting period of the first delay period $t_{sub.dl_1}$, wherein the first delay period $t_{sub.dl_1}$ is different for each memory die. By imposing a unique time delay for each memory die, memory dies can enter the subsequent operation step **S1030** one at a time. The memory dies completing the operation step **S1020** at the same time, can be desynchronized at the operation step **S1025**. Similar to the operation step **S625**, the operation step **S1025** ensures that only one additional peak power operation can be allowed at any given time.

(110) The implementation of operation steps **S1030**, **S1035** and **S1040** are similar to operation steps **S630**, **S635** and **S640** in FIG. 6. When the PPM enablement signal $\text{enPPM2}=0$ at operation step **S1030**, the PPC routine **1000** moves to operation step **S1045**, where the pull-up driver **314** of the PPM2 circuit **206-1** can be switched on for the NAND flash memory **100-1**.

(111) At operation step **S1050**, the NAND flash memory **100-1** starts to perform the peak power operation with the peak current $I_{\text{sub.cc}}$. If the maximum number of peak power operations have been reached after the NAND flash memory **100-1** starts the peak power operation, the electric potential $V_{\text{sub.ppm}}$ of the PPM2 group can be increased to equal or above the reference voltage $V_{\text{sub.ref}}$ and thereby the PPM enablement signal enPPM2 will be set to 1 (i.e., $\text{enPPM2}=1$). If the maximum number of peak power operations have not been reached after the NAND flash memory **100-1** starts the peak power operation, the electric potential $V_{\text{sub.ppm}}$ of the PPM2 group is still lower than the reference voltage $V_{\text{sub.ref}}$. The PPM enablement signal enPPM2 remains as zero (i.e., $\text{enPPM2}=0$).

(112) At operation step **S1055**, after completing the peak power operation, the pull-up driver **314** of the PPM2 circuit **206-1** can be disabled (i.e., switched off) for the NAND flash memory **100-1**. Accordingly, the PPM enablement signal enPPM2 for the PPM2 group can be reset to zero again.

(113) At operation step **S1060**, the PPC routine **1000** performs the second delay period $t_{\text{sub.dl}_2}$, similar to the operation step **S660** in FIG. 6.

(114) At operation step **S1065**, the pull-up driver **314** of the PPM1 circuit **202-1** on the NAND flash memory **100-1** is disabled (i.e., switched off). Accordingly, the PPM enablement signal enPPM1 for the PPM1 group can be reset to zero again. Another batch of multiple memory dies can subsequently pass the first check point at operation step **S1010** and are queued for peak power operations.

(115) FIG. 11 illustrates an exemplary PPM circuit **1100** on the NAND flash memory **100**, according to some embodiments of the present disclosure, where each of the PPM1 circuits **202** and each of the PPM2 circuits **206** shown in FIG. 2 can be similar to the PPM circuit **1100**. Structurally, the PPM circuit **1100** is similar to the PPM circuit **300** in FIG. 3 except that 1) the pull-up current $I_{\text{sub.pull-up}}$ controlled by the pull-up driver **314** is used to signal the peak power operation performed on the memory die instead of the pull-down current $I_{\text{sub.pull-dn}}$ in FIGS. 3 and 2) the inverter **332** of the output circuit in FIG. 3 can be removed. In this example, the pull-up driver **314** can be switched on or off to control the pull-up current $I_{\text{sub.pull-up}}$ flowing through the pull-up driver **314** and the pull-up resistor **318**. The peak current $I_{\text{sub.cc}}$ of a memory die (e.g., the NAND flash memory **100-1**) used during the peak power operation can correspond to the voltage level of the first control signal **340**, which can switch on the pull-up driver **314**.

(116) The operation (i.e., switching on/off pull-up and pull-down drivers **314/366**) of the PPM circuit **1100** is similar to the PPM circuit **700** in FIG. 7 where the difference lies in the relationship between the electric potential $V_{\text{sub.ppm}}$ and the number m of peak power operations. In this example, one pull-down driver **336** can be selected to be switched on (i.e., enabled) within each PPM group (e.g., within the PPM1 group **210** or the PPM2 group **211** in FIG. 2) during operation and the other pull-down drivers **336** in the same PPM group can be switched off. As such, in each PPM group, current only flows from the pull-down resistor **320** through the pull-down driver **336**, which is switched on, to the ground. During operation, the pull-up driver **314** can be switched on or off depending on the status of the memory die, and can be independently controlled according to the PPM management scheme, similar to that of the PPM circuit **700** described with respect to FIGS. 7-10, where the differences will be discussed below.

(117) FIG. 12 shows an equivalent circuit **1200** of a PPM group using the PPM circuits **1100** shown in FIG. 11, according to some embodiments of the present disclosure. Similar to the equivalent circuit shown in FIG. 8, only one pull-down driver **336** is switched on in one PPM group for peak power management, and the pull-up drivers **314** (e.g., **314-1**, **314-2**, . . . , **314-m**) can be switched on according to the memory die performing the peak power operation, i.e., using the peak current

I.sub.cc. The difference between FIG. 8 and FIG. 12 is that the pull-up current sources 714 is used in FIG. 8 and the pull-up resistors 318 (e.g., 318-1, 318-2, . . . , 318-m) is used in FIG. 12. The electric potential V.sub.ppm of the node 322 can be expressed as a function of m, the number of peak power operations, i.e., the number of memory dies performing peak power operations if one memory die performs one peak power operation:

$$(118) 0 V_{ppm}(m) = \frac{mR_d}{mR_d + R_u} V_{dd} \quad (6)$$

where R.sub.d and R.sub.u are the resistances of the pull-down resistor 320 and the pull-up resistor 318, respectively. In some embodiments, the magnitude of the pull-up current I.sub.pull-up (e.g., I.sub.pull_up-1, I.sub.pull_up-2, . . . , I.sub.pull_up-m) can be different on different memory dies and a person of skill in the art can modify the Equation (6) accordingly. As shown in Equation (6), the electrical potential V.sub.ppm at the node 322 (or at the first and second PPM contact pads 204/208) increases when the number m of the peak power operations increases. As the number of peak power operations (i.e., the variable m) changes continuously, the electric potential V.sub.ppm changes continuously accordingly. As such, through an analog circuit like the PPM circuit 1100, information about the peak power operations in each PPM group can be encoded into the electric potential V.sub.ppm of the PPM pins in an analog way.

(119) FIG. 13 illustrates an exemplary peak power management scheme using the PPM circuit 1100 (FIG. 11, see also equivalent circuit 1200 in FIG. 12), according to some embodiments of the present disclosure. In this example, it is assumed that the resistance R.sub.u is twice as much as the resistance R.sub.d and the voltage V.sub.dd of the power source 312 is 1.2V.

(120) Referring to FIGS. 12 and 13, to start peak power management, any one of the pull-down driver 336 in the PPM group can be switched on. If there is no memory die performing peak power operation, none of the pull-up drivers 314 is switched on. The electric potential V.sub.ppm of the node 322 is connected to the ground through the conductive path established by the pull-down resistor 320 and the specific pull-down driver 336 that is switched on. As such, V.sub.ppm(0)=0 V(at m=0).

(121) When one of the memory dies, e.g., the NAND flash memory 100-1, is performing the peak power operation, the memory die consumes the peak current I.sub.cc-1. The first control signal 340-1 can be generated to switch on the pull-up driver 314-1 of the PPM circuit on the NAND flash memory 100-1 to signal the peak power operation on the NAND flash memory 100-1. The pull-up current I.sub.pull_up-1 can flow from the power source 312 through the pull-up driver 314-1, the pull-up resistor 318-1, the pull-down resistor 320 and the pull-down driver 336 to the ground. The electric potential V.sub.ppm at the node 322 is changed to

$$(122) V_{ppm}(1) = \frac{R_d}{R_d + R_u} V_{dd} = 0.4V(at m = 1).$$

(123) When another memory die, e.g., the NAND flash memory 100-2, is also performing the peak power operation, the first control signal 340-2 can be generated to switch on the pull-up driver 314-2 to signal the peak power operation on the NAND flash memory 100-2. As such, the pull-up current I.sub.pull_up-2 can flow through the pull-up driver 314-2 and the pull-up resistor 318-2. The pull-up current I.sub.pull_dn-1 and the pull-down current I.sub.pull_dn-2 flow through the pull-down resistor 320 and the pull-down driver 336 as the pull-down current I.sub.pull_dn. The electric potential V.sub.ppm at the node 322 is then changed to

$$(124) V_{ppm}(2) = \frac{2R_d}{2R_d + R_u} V_{dd} = 0.6V(at m = 2).$$

(125) Similarly, when a third memory die, e.g., the NAND flash memory 100-3, is also performing the peak power operation, the electric potential V.sub.ppm at the node 322 is then changed to

$$(126) V_{ppm}(3) = \frac{3R_d}{3R_d + R_u} V_{dd} = 0.72V(at m = 3).$$

Using the Equation (6), the electric potential V.sub.ppm can be calculated if four memory dies (m=4) in the storage system 10 are using the peak current I.sub.cc, i.e. performing the peak power operations. The electric potential

$$(127) V_{ppm}(4) = \frac{4R_d}{4R_d + R_u} V_{dd} = 0.8V.$$

(128) The electric potentials $V_{\text{sub.ppm}}(m)$ are listed in FIG. 13 for various number of peak power operations (i.e., 0, 1, 2, 3, 4, . . . , $m-1$, m), where m is a positive integer.

(129) As shown in FIG. 13, the electric potential $V_{\text{sub.ppm}}$ increases when the number of the peak power operations increases. Therefore, the reference voltage $V_{\text{sub.ref}}$ for the comparator 328 (in FIG. 11) can be chosen such that the PPM enablement signal enPPM at the output terminal 330 of the comparator 328 can be set at, for example, logic zero ($\text{enPPM}=0$) when the electric potential $V_{\text{sub.ppm}}$ is lower than the reference voltage $V_{\text{sub.ref}}$, i.e., when the number of memory dies performing peak power operations is less than a predetermined number. And the PPM enablement signal enPPM can be set at $\text{enPPM}=1$ when the electric potential $V_{\text{sub.ppm}}$ is not lower than the reference voltage $V_{\text{sub.ref}}$, i.e., when the number of memory dies performing peak power operations has reached (equal or greater than) the predetermined number.

(130) Similar to the embodiment shown in FIG. 7, the reference voltage $V_{\text{sub.ref}}(m)$, corresponding to the maximum number m of peak power operations, can be set to a voltage in a range between the electric potential $V_{\text{sub.ppm}}(m)$ and the electric potential $V_{\text{sub.ppm}}(m-1)$:

$$(131) \quad \frac{(m-1)R_d}{(m-1)R_d + R_u} V_{dd} < V_{\text{ref}}(m) < \frac{mR_d}{mR_d + R_u} V_{dd} \quad (7)$$

(132) In the example shown in FIG. 13, if the maximum number of peak power operations allowed is 2, the reference voltage $V_{\text{sub.ref}}$ can be a voltage in a range between $V_{\text{sub.ppm}}(2)=0.6\text{V}$ and $V_{\text{sub.ppm}}(1)=0.4\text{V}$, where $V_{\text{sub.ref}}(2)=0.5\text{V}$ can be chosen in FIG. 13 as an example. If the electric potential $V_{\text{sub.ppm}}$ is less than the reference voltage $V_{\text{sub.ref}}=0.5\text{V}$, the PPM enablement signal $\text{enPPM}=0$ is generated at the output terminal 330 (see FIG. 11), indicating that the number of peak power operations is 0 or 1, less than the maximum number of 2. The storage system can have at least one additional peak power operation performed by at least one of the memory dies. When there are two peak power operations performed simultaneously, e.g., two memory dies are consuming the peak current $I_{\text{sub.cc}}$, the electric potential $V_{\text{sub.ppm}}=0.6\text{V}$, higher than the reference voltage $V_{\text{sub.ref}}=0.5\text{V}$. The PPM enablement signal $\text{enPPM}=1$ can be generated accordingly, indicating that the maximum number of peak power operations of 2 has been reached. The NAND storage system can then prohibit additional memory die to perform a peak power operation. As such, the PPM enablement signal enPPM can function as an indicator for the number of peak power operations performed in the storage system with multiple memory dies. Through analog PPM circuit 1100 and voltages (e.g., analog voltages) associated with the electric potential $V_{\text{sub.ppm}}$, the reference voltage $V_{\text{sub.ref}}$, the first and second control signals 340/342, etc., multiple peak power operations can be controlled simultaneously.

(133) With respect to algorithm for managing peak power operations using the PPM circuits 1100, the peak power check routine can be very similar to that described in FIG. 10.

(134) FIG. 14 illustrates an exemplary PPM circuit 1400 on the NAND flash memory 100, according to some embodiments of the present disclosure, where each of the PPM1 circuits 202 and each of the PPM2 circuits 206 shown in FIG. 2 can be similar to the PPM circuit 300. The PPM circuit 1400 is similar to the PPM circuit 300 except that the pull-down resistor 320 in FIG. 3 can be replaced by a pull-down current source 1436.

(135) The pull-down current source 1436 can be any current source that provide constant current. In some embodiments, the pull-down current source 1436 can be a transistor, for example, a field-effect transistor (FET). In FIG. 14, a n-channel MOSFET is shown as an example. However, the pull-down current source 1436 is not so limiting. For example, the pull-down current source 1436 can be a bipolar transistor. In FIG. 14, the pull-down driver 336 is also a n-channel MOSFET. However, the pull-down driver 336 is not so limiting. The pull-down driver 336 can also be any suitable transistor. The functionality of the other components of the PPM circuit 1400 is similar to the PPM circuit 300 described with respect to FIG. 3, and only the differences will be discussed below.

(136) In FIG. 14, the magnitude of the pull-down current $I_{\text{sub.pull_dn}}$ flowing through the pull-

pull-down current source **1436** can be controlled through a gate terminal **1438**. The pull-down current $I_{sub.pull_dn}$ can be determined by the voltage level at the gate terminal **1438** and the transconductance of the pull-down current source **1436**. In this example, the pull-down driver **336** can be switched on or off to signal whether there is a peak power operation performed on the memory die. For example, the pull-down driver **336** can be switched on when a voltage applied to the gate terminal **338** of the pull-down driver **336** corresponds to the peak current $I_{sub.cc}$. As such, the peak power operations can be regulated for each memory die through the pull-down current $I_{sub.pull_dn}$. Similar to the PPM circuit **300** in FIG. 3, through the die-to-die connections at the PPM contact pads, peak power operations throughout the entire storage system can be coordinated between different memory dies using the PPM circuit **1400**.

(137) Similar to the operations of the PPM circuits **300** in FIGS. 3-6, implementation of the PPM circuits **1400** can include by switching on only one pull-up driver **314** within each PPM group (e.g., within the PPM1 group **210** or the PPM2 group **211** in FIG. 2) for peak power management and by switching off the other pull-up drivers **314** in the same PPM group. As such, current only flows from the power source **312** through one pull-up resistor **318** in each PPM group. During operation, the pull-down driver **336** can be switched on or off depending on the status of the memory die, and can be independently controlled according to the PPM management scheme discussed below. Accordingly, in each PPM group, the electric potential $V_{sub.ppm}$ of the node **322** (or the PPM pins) depends on the number of pull-down drivers **336** that are switched on in the respective PPM group. By monitoring the electric potential $V_{sub.ppm}$, the number of peak power operations performed in a storage system having multiple memory dies can thereby be regulated.

(138) FIG. 15 shows an equivalent circuit **1500** of a PPM group, according to some embodiments of the present disclosure. The PPM group can be the PPM1 group **210** or the PPM2 group **211** in the storage system in FIG. 2. The pull-up drivers **314** and pull-down drivers **336** that are switched off can be omitted from the equivalent circuit **1500**. As discussed previously, the node **322** can be held at the same electrical potential $V_{sub.ppm}$ in the entire PPM group between different memory dies.

(139) In some embodiments, only one pull-up driver **314** is switched on in one PPM group for peak power management. In some embodiments, the pull-down driver **336** is switched on in the PPM circuit corresponding to the memory die performing the peak power operation, i.e., using the peak current $I_{sub.cc}$. The pull-down drivers **336** in the same PPM group are connected in parallel with each other. In this configuration, the pull-up current $I_{sub.pull_up}$ flowing through the pull-up driver **314** that is switched on, is the sum of the pull-down current $I_{sub.pull_dn}$ (e.g., $I_{sub.pull_dn-1}$, $I_{sub.pull_dn-2}$, . . . , $I_{sub.pull_dn-m}$) flowing through each of the pull-down drivers **336** that are switched on. The electric potential $V_{sub.ppm}$ of the node **322** can be expressed as a function of m , the number of peak power operations, i.e., the number of memory dies performing peak power operations if one memory die performs one peak power operation:

$$V_{sub.ppm}(m) = V_{sub.dd} - m \cdot i \cdot R_{sub.u} \quad (8)$$

(140) where $V_{sub.dd}$ is the voltage of the power source, $R_{sub.u}$ is the resistances of the pull-up resistor **318**, and i is the magnitude of the pull-down current $I_{sub.pull_dn}$ controlled by the pull-down current sources **1436** (e.g., **1436-1**, **1436-2**, . . . , **1436-m**). In some embodiments, the magnitude of the pull-down current $I_{sub.pull_dn}$ (e.g., $I_{sub.pull_dn-1}$, $I_{sub.pull_dn-2}$, . . . , $I_{sub.pull_dn-m}$) can be different on different memory dies and a person of skill in the art can modify the Equation (8) accordingly. As shown in Equation (8), by using the pull-down current source **1436** in FIG. 14 instead of the pull-up resistor **318** in FIG. 3, the electrical potential $V_{sub.ppm}$ at the node **322** (or at the first and second PPM contact pads **204/208**) can have a liner relationship with the number m of the peak power operations. In this example, the electrical potential $V_{sub.ppm}$ decreases when the number m of the peak power operations increases. As the number of peak power operations (i.e., the variable m) changes continuously, the electric potential $V_{sub.ppm}$ changes continuously accordingly. As such, through an analog circuit like the PPM

circuit **1400**, information about the peak power operations in each PPM group can be encoded into the electric potential $V_{\text{sub.ppm}}$ of the PPM pins in an analog way.

(141) FIG. **16** illustrates an exemplary peak power management scheme, according to some embodiments of the present disclosure. In this example, the voltage $V_{\text{sub.dd}}$ of the power source **312** is assumed to be 1.2 V. The resistance $R_{\text{sub.u}}$ of the pull-up resistor **318** is assumed to be 1000 ohm. The magnitude i of the pull-down current $I_{\text{sub.pull_dn}}$ is 160 μA . It is noted that the assumptions of the voltage $V_{\text{sub.dd}}$, the resistance $R_{\text{sub.u}}$ and the magnitude i of the pull-down current $I_{\text{sub.pull_dn}}$ are chosen to provide an illustration of the principles of the peak power management scheme in the present disclosure. A person of ordinary skill in the art can utilize the principle of the peak power management scheme described here with various modifications as suitable for the practical applications. All such modifications and variations are within the scope of the present disclosure.

(142) At the beginning of peak power management, any one of the pull-up driver **314** in the PPM group can be switched on. If there is no memory die performing peak power operation, none of the pull-down drivers **336** is switched on. The electric potential $V_{\text{sub.ppm}}$ of the node **322** is set to the voltage $V_{\text{sub.dd}}$ of the power source **312** through the conductive path established by the specific pull-up driver **314** that is switched on. If $V_{\text{sub.dd}}=1.2\text{V}$, then $V_{\text{sub.ppm}}(0)=1.2\text{ V}$ (at $m=0$).

(143) When one of the memory dies, e.g., the NAND flash memory **100-1** in FIG. **2**, is performing the peak power operation, the memory die consumes the peak current $I_{\text{sub.cc-1}}$. As discussed previously, the second control signal **342-1** can be generated to switch on the pull-down driver **336-1** to enable the peak power operation on the NAND flash memory **100-1**. As such, the pull-down current i can flow through the pull-down driver **336-1** and the pull-down current source **1436-1**. The electric potential $V_{\text{sub.ppm}}$ at the node **322** is changed to $V_{\text{sub.ppm}}(1)=V_{\text{sub.dd}}-iR_{\text{sub.u}}=1.04\text{ V}$ (at $m=1$).

(144) When another memory die, e.g., the NAND flash memory **100-2** (in FIG. **2**), is also performing the peak power operation, there are two memory dies ($m=2$) in the storage system **10** are now using the peak current $I_{\text{sub.cc}}$, i.e., the peak current $I_{\text{sub.cc-1}}$ on the NAND flash memory **100-1** and the peak current $I_{\text{sub.cc-2}}$ on the NAND flash memory **100-2**. As discussed previously, the second control signal **342-2** can be generated to switch on the pull-down driver **336-2** to enable the peak power operation on the NAND flash memory **100-2**. As such, the pull-down current $I_{\text{sub.pull_dn-2}}$ can flow through the pull-down driver **336-2** and the pull-down current source **1436-2**. The pull-down current $I_{\text{sub.pull_dn-1}}$ and the pull-down current $I_{\text{sub.pull_dn-2}}$ flow through the pull-down current source **1436-1** and the pull-down current source **1436-2**, respectively, where the pull-down current source **1436-1** and the pull-down current source **1436-2** are connected in parallel. The electric potential $V_{\text{sub.ppm}}$ at the node **322** is then changed to $V_{\text{sub.ppm}}(2)=V_{\text{sub.dd}}-2iR_{\text{sub.u}}=0.88\text{ V}$ (at $m=2$).

(145) When a third memory die, e.g., the NAND flash memory **100-3**, is also performing the peak power operation, i.e., three memory dies ($m=3$) in the storage system **10** are consuming the peak current $I_{\text{sub.cc}}$, the electric potential $V_{\text{sub.ppm}}$ at the node **322** is then changed to $V_{\text{sub.ppm}}(3)=V_{\text{sub.dd}}-3iR_{\text{sub.u}}=0.72\text{ V}$ (at $m=3$).

(146) Using the Equation (8), the electric potential $V_{\text{sub.ppm}}$ can be calculated if four memory dies ($m=4$) in the storage system **10** are using the peak current $I_{\text{sub.cc}}$, i.e. performing the peak power operations. The electric potential $V_{\text{sub.ppm}}(4)=V_{\text{sub.dd}}-4iR_{\text{sub.u}}=0.56\text{ V}$. The electric potentials $V_{\text{sub.ppm}}(m)$ are listed in FIG. **16** for various number of peak power operations (i.e., 0, 1, 2, 3, 4, . . . , $m-1$, m).

(147) As discussed above, the electric potential $V_{\text{sub.ppm}}$ of a PPM group depends on the number of peak power operations. According to Equation (8) and FIG. **16**, the electric potential $V_{\text{sub.ppm}}$ decreases when the number of the peak power operations increases. Therefore, the reference voltage $V_{\text{sub.ref}}$ for the comparator **328** (in FIG. **14**) can be chosen such that the PPM enablement signal enPPM can be set at $\text{enPPM}=0$ when the electric potential $V_{\text{sub.ppm}}$ is higher than the

reference voltage $V_{sub.ref}$, i.e., when the number of memory dies performing peak power operations is less than a predetermined number. And the PPM enablement signal $enPPM$ can be set at $enPPM=1$ when the electric potential $V_{sub.ppm}$ is not higher than the reference voltage $V_{sub.ref}$, i.e., when the number of memory dies performing peak power operations has reached (equal or greater than) the predetermined number. In the other words, the reference voltage $V_{sub.ref}$ can be programmed to correspond to the maximum number of peak power operations allowed in a storage system. Accordingly, when the PPM enablement signal $enPPM=0$, the storage system can still provide additional peak current $I_{sub.cc}$ to at least one of the memory dies, i.e., having enough power (or current) to provide at least one additional memory die to perform peak power operation. On the contrary, when the PPM enablement signal $enPPM=1$, the maximum number of peak power operations have been reached for the storage system. The storage system has reached its power (or current) limit and cannot provide additional peak current $I_{sub.cc}$ to any of the memory dies to perform any additional peak power operation.

(148) For a storage system where each memory die includes more than one PPM group, one reference voltage $V_{sub.ref}$ can be chosen for each PPM group to output a PPM enablement signal $enPPM$ for controlling a maximum number of peak power operations. In the example of the storage system in FIG. 2, the PPM1 group can have a reference voltage $V_{sub.ref-1}$ programed to correspond to a number $m_{sub.1}$ —the maximum number of peak power operations allowed in the PPM1 group. Similarly, the PPM2 group can have a reference voltage $V_{sub.ref-2}$ programed to correspond to a number $m_{sub.2}$ —the maximum number of peak power operations allowed in the PPM2 group.

(149) For illustration purpose below, it is assumed that the maximum number of peak power operations in a storage system (or in each memory group) is the number m , and each memory die can perform a peak power operation at a time. It is noted that these assumptions are chosen for simplicity and should not limit the scope of this disclosure. For example, each memory die can perform two or more peak power operations simultaneously.

(150) According to Equation (8), if there are $m-1$ number of peak power operations, the electric potential $V_{sub.ppm}$ can be expressed as:

$$V_{sub.ppm}(m-1)=V_{sub.dd}-(m-1)*i*R_{sub.u} \quad (9)$$

Thus, the reference voltage $V_{sub.ref}(m)$, corresponding to the maximum number m of peak power operations, can be set to a voltage in a range between the electric potential $V_{sub.ppm}(m)$ and the electric potential $V_{sub.ppm}(m-1)$:

$$V_{sub.dd}-m*i*R_{sub.u}<V_{sub.ref}(m)<V_{sub.dd}-(m-1)*i*R_{sub.u} \quad (10)$$

(151) In the example shown in FIG. 16, if the maximum number of peak power operations allowed is 2, the reference voltage $V_{sub.ref}$ can be a voltage in a range between $V_{sub.ppm}(2)=0.88$ V and $V_{sub.ppm}(1)=1.12$ V, where $V_{sub.ref}(2)=0.96$ V is chosen in FIG. 16 as an example. If the electric potential $V_{sub.ppm}$ is higher than the reference voltage $V_{sub.ref}=0.96$ V, the PPM enablement signal $enPPM=0$ is generated at the output terminal 334 (see FIG. 14), indicating that the number of peak power operations is 0 or 1, less than the maximum number of 2. The storage system can have at least one additional peak power operation performed by at least one of the memory dies. When there are two peak power operations performed simultaneously, i.e., two memory dies are consuming the peak current $I_{sub.cc}$, the electric potential $V_{sub.ppm}=0.88$ V, lower than the reference voltage $V_{sub.ref}=0.96$ V. The PPM enablement signal $enPPM=1$ is generated accordingly, indicating that the maximum number of peak power operations of 2 has been reached. The NAND storage system can then prohibit additional memory die to perform a peak power operation. As such, the PPM enablement signal $enPPM$ can function as an indicator for the number of peak power operations performed in the storage system with multiple memory dies.

(152) As discussed above, the PPM enablement signal $enPPM$ for each PPM group can be generated by comparing the reference voltage $V_{sub.ref}$ with the electric potential $V_{sub.ppm}$ of the PPM pins (or at the node 322) that are electrically connected between the memory dies. The PPM

enablement signal enPPM is set to 0 if the electric potential $V_{\text{sub.ppm}}$ of the PPM pins is higher than the reference voltage $V_{\text{sub.ref}}$. The PPM enablement signal enPPM is set to 1 if the electric potential $V_{\text{sub.ppm}}$ of the PPM pins is not higher than the reference voltage $V_{\text{sub.ref}}$. Because the electric potential $V_{\text{sub.ppm}}$ depends on the pull-down current $I_{\text{sub.pull_dn}}$ flowing through each of the pull-down drivers in the PPM group, and the pull-down driver is switched on when peak power operation is performed on the corresponding memory die, thus the reference voltage can be selected based on a maximum number of peak power operations for the storage system. Through analog PPM circuit **1400** and voltages (e.g., analog voltages) associated with the electric potential $V_{\text{sub.ppm}}$, the reference voltage $V_{\text{sub.ref}}$, gate voltage applied to the pull-down current sources **1436**, etc., multiple peak power operations can be controlled simultaneously.

(153) With respect to algorithm for managing peak power operations using the PPM circuits **1400**, the peak power check routine can be very similar to that described in FIG. **6** with similar equivalent circuit.

(154) In summary, all of the above analog peak power management methods can be applied to an asynchronous multi-plane independent (AMPI) read operation. In a NAND storage system, each storage system includes multiple memory dies and each memory die can have multiple memory planes. For example, the NAND flash memory **100** shown in FIG. **1B** includes four memory planes **101**. Read operations from different memory planes need to be staggered to avoid coincidence of peak current, and thereby are referred to as AMPI read. In some embodiments, an AMPI read can be defined as a peak power operation. Each memory plane's AMPI can be controlled by a PPM circuit, or more specifically can be controlled through the pull-down driver. The PPM circuits from a group of memory planes can be electrically connected through the PPM pins to form a PPM group, where the maximum number of peak power operations can be predetermined according to the peak power (or peak current) budget of the NAND storage system.

(155) The present disclosure also provides a method of peak power management (PPM) for a storage system with multiple memory dies, wherein each of the multiple memory dies includes a first PPM circuit and a second PPM circuit. First PPM circuits of the multiple memory dies are electrically connected to form a first PPM group, and second PPM circuits of the multiple memory dies are electrically connected to form a second PPM group. The method includes the following steps: switching on a first pull-down driver of the first PPM circuit on a selected memory die when a first PPM enablement signal of the first PPM group is zero; waiting for a first delay period; switching on a second pull-down driver of the second PPM circuit on the selected memory die when a second PPM enablement signal of the second PPM group is zero; and performing a peak power operation on the selected memory die. The first and second PPM enablement signals depend on the current flowing through each of the pull-down drivers in the first and second PPM groups.

(156) The present disclosure provides a peak power management (PPM) circuit for a storage system with multiple memory dies. The PPM circuit includes a pull-up driver electrically connected between a power source and a PPM pin; and a pull-down driver electrically connected between a ground and the PPM pin, wherein the pull-up driver and the pull-down driver are connected in series; and the PPM circuit is configured to manage a peak power operation for the memory die based on an electric potential of the PPM pin, wherein PPM pins of PPM circuits on different memory dies are electrically connected.

(157) The present disclosure discloses a method of peak power management (PPM) for a storage system with multiple memory dies, wherein each of the multiple memory dies comprises a PPM circuit, and each PPM circuit comprises a pull-up driver connected in series with a pull-down driver. The method includes sending a first control signal to switch on a corresponding pull-up driver on one of the multiple memory dies; sending a second control signal to switch on a respective pull-down driver on a selected memory die based on a PPM enablement signal; and performing a peak power operation on the selected memory die.

(158) The present disclosure further discloses a method of peak power management (PPM) for a

storage system with multiple memory dies, wherein each of the multiple memory dies comprises a PPM circuit, and each PPM circuit comprises a pull-up driver connected in series with a pull-down driver. The method includes sending a second control signal to switch on a corresponding pull-down driver on one of the multiple memory dies; sending a first control signal to switch on a respective pull-up driver on a selected memory die based on a PPM enablement signal; and performing a peak power operation on the selected memory die.

(159) The foregoing description of the specific embodiments will so fully reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt, for various applications, such specific embodiments, without undue experimentation, and without departing from the general concept of the present disclosure.

Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the disclosure and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the disclosure and guidance.

(160) Embodiments of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

(161) The Summary and Abstract sections can set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

(162) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A peak power management (PPM) circuit for a storage system with multiple memory dies, the PPM circuit being provided in a memory die of the multiple memory dies and comprising: a PPM pad, wherein PPM pads of PPM circuits including at least the PPM circuit in the multiple memory dies are electrically connected; a pull-up driver electrically connected between a power source and the PPM pad; a pull-down driver electrically connected between a ground and the PPM pad, wherein: the pull-up driver and the pull-down driver are connected in series; and one of the pull-up driver and the pull-down driver is controlled by a control signal corresponding to a peak power operation performed on the memory die so that an electric potential of the PPM pad is determined by the peak power operation in an analog way; and a filter coupled to the PPM pad.
2. The PPM circuit of claim 1, further comprising: a pull-up resistor electrically connected between the power source and the PPM pad, wherein the pull-up resistor is connected in series with the pull-up driver.
3. The PPM circuit of claim 1, further comprising: a pull-down resistor electrically connected between the ground and the PPM pad, wherein the pull-down resistor is connected in series with the pull-down driver.
4. The PPM circuit of claim 1, further comprising: a pull-up current source electrically connected between the power source and the PPM pad, wherein: the pull-up current source is connected in series with the pull-up driver and is configured to provide constant current; and the pull-up driver is controlled by the control signal corresponding to the peak power operation.
5. The PPM circuit of claim 1, further comprising: a pull-down current source electrically connected between the ground and the PPM pad, wherein: the pull-down current source is

connected in series with the pull-down driver and is configured to provide constant current; and the pull-down driver is controlled by the control signal corresponding to the peak power operation.

6. The PPM circuit of claim 1, further comprising: an output circuit for outputting a PPM enablement signal, wherein the PPM pad is electrically connected through the filter with a first input terminal of the output circuit, and the output circuit is configured to output the PPM enablement signal according to the electric potential of the PPM pad.

7. The PPM circuit of claim 6, wherein the output circuit comprises: a comparator with a first input terminal electrically connected to the PPM pad through the filter and a second input terminal electrically connected to a reference voltage.

8. The PPM circuit of claim 7, wherein the reference voltage is based on a maximum number of peak power operations allowed for the multiple memory dies.

9. The PPM circuit of claim 8, wherein the PPM enablement signal is set as a first logic state when the electric potential of the PPM pad is not higher than the reference voltage, and is set as a second logic state when the electric potential of the PPM pad is higher than the reference voltage.

10. The PPM circuit of claim 7, wherein the output circuit further comprises: an inverter with an input connected to an output terminal of the comparator, wherein the inverter is configured to output the PPM enablement signal for determining whether to perform the peak power operation.

11. The PPM circuit of claim 1, wherein the electric potential of the PPM pad depends on a number of peak power operations performed by the multiple memory dies in the storage system.

12. A peak power management (PPM) system for a storage system with multiple memory dies, comprising one or more PPM groups, each of the one or more PPM groups comprising multiple PPM circuits, each of the multiple PPM circuits being provided in a memory die and comprising: a PPM pad, wherein in each of the one or more PPM groups, PPM pads of PPM circuits in different memory dies are electrically connected; a pull-up driver electrically connected between a power source and the PPM pad; a pull-down driver electrically connected between a ground and the PPM pad; and a filter coupled to the PPM pad, wherein the PPM system is configured to manage a peak power operation for the multiple memory dies based on an electric potential of the PPM pad.

13. The PPM system of claim 12, wherein in each of the multiple PPM circuits, the pull-up driver and the pull-down driver are connected in series, and one of the pull-up driver and the pull-down driver is controlled by a control signal corresponding to the peak power operation performed on the memory die so that the electric potential of the PPM pad is determined by the peak power operation in an analog way.

14. The PPM system of claim 13, wherein the electric potential of the PPM pad in each of the one or more PPM groups depends on a number of peak power operations performed by the multiple memory dies in the storage system.

15. The PPM system of claim 13, wherein each of the multiple PPM circuits further comprises: a pull-up current source electrically connected between the power source and the PPM pad, wherein: the pull-up current source is connected in series with the pull-up driver and is configured to provide constant current; and the pull-up driver is controlled by the control signal corresponding to the peak power operation.

16. The PPM system of claim 15, wherein the electric potential of the PPM pad in each of the one or more PPM groups increases linearly with a number of peak power operations performed by the multiple memory dies of the storage system.

17. The PPM system of claim 13, wherein each of the multiple PPM circuits further comprises: a pull-down current source electrically connected between the ground and the PPM pad, wherein: the pull-down current source is connected in series with the pull-down driver and is configured to provide constant current; and the pull-down driver is controlled by the control signal corresponding to the peak power operation.

18. The PPM system of claim 17, wherein the electric potential of the PPM pad in each of the one or more PPM groups decreases linearly with a number of peak power operations performed by the

multiple memory dies of the storage system.

19. The PPM system of claim 12, wherein each of the multiple PPM circuits further comprises: a pull-up resistor electrically connected between the power source and the PPM pad, wherein the pull-up resistor is connected in series with the pull-up driver.

20. The PPM system of claim 12, wherein each of the multiple PPM circuits further comprises: a pull-down resistor electrically connected between the ground and the PPM pad, wherein the pull-down resistor is connected in series with the pull-down driver.

21. The PPM system of claim 12, wherein each of the one or more PPM groups further comprises: an output circuit for outputting a PPM enablement signal, wherein the PPM pads of the PPM circuits of a respective PPM group are electrically connected with a first input terminal of the output circuit through the filter, and the output circuit is configured to output the PPM enablement signal according to the electric potential of the PPM pads.

22. The PPM system of claim 21, wherein the output circuit comprises: a comparator with a first input terminal electrically connected to the PPM pads of the PPM circuits of the respective PPM group through the filter and a second input terminal electrically connected to a reference voltage.

23. The PPM system of claim 22, wherein the reference voltage is based on a maximum number of peak power operations allowed for the multiple memory dies.

24. The PPM system of claim 23, wherein the PPM enablement signal is set as a first logic state when the electric potential of the PPM pad is not higher than the reference voltage, and is set as a second logic state when the electric potential of the PPM pad is higher than the reference voltage.

25. The PPM system of claim 22, wherein the output circuit further comprises: an inverter with an input connected to an output terminal of the comparator, wherein the inverter is configured to output the PPM enablement signal for determining whether to perform the peak power operation.

26. The PPM system of claim 12, wherein pull-up drivers in different memory dies are connected in parallel with each other in each of the one or more PPM groups when each of the pull-up drivers is controlled by a respective control signal corresponding to a respective peak power operation performed by a respective memory die; and wherein only one pull-down driver in each of the one or more PPM groups is switched on.

27. The PPM system of claim 12, wherein pull-down drivers in different memory dies are connected in parallel with each other in each of the one or more PPM groups when each of the pull-down drivers is controlled by a respective control signal corresponding to a respective peak power operation performed by a respective memory die; and wherein only one pull-up driver in each of the one or more PPM groups is switched on.

28. A method of peak power management (PPM) for a storage system with multiple memory dies, wherein each of the multiple memory dies comprises a PPM circuit, and each PPM circuit comprises a pull-up driver connected in series with a pull-down driver, the method comprising: receiving a first control signal to switch on a first pull-up driver or a first pull-down driver on one of the multiple memory dies; receiving a second control signal to switch on a second pull-down driver or a second pull-up driver on a certain memory die based on a PPM enablement signal; generating the PPM enablement signal by comparing a reference voltage with an electric potential of a PPM pad of the PPM circuit, wherein PPM pads of PPM circuits on different memory dies are electrically connected and the electric potential depends on a number of peak power operations performed by the multiple memory dies in the storage system; and performing a peak power operation on the certain memory die.

29. The method of claim 28, wherein the receiving of the second control signal comprises applying an analog voltage on a gate terminal of the second pull-down driver or the second pull-up driver.

30. The method of claim 29, further comprising: switching on the second pull-down driver or the second pull-up driver based on a peak power current used during the peak power operation.

31. The method of claim 28, further comprising: selecting the reference voltage based on a maximum number of peak power operations for the storage system.

32. The method of claim 28, further comprising: generating, by a pull-down current source or a pull-up current source, a pull-down current or a pull-up current, respectively, which is proportional to a peak power current for the peak power operation.
33. A memory die, comprising: a communication interface; a PPM circuit coupled to the communication interface, the PPM circuit comprising: a PPM pad, wherein the PPM pad is electrically connected to the PPM pad of the PPM circuit of at least one other memory die of multiple memory dies including the memory die; a pull-up driver electrically connected between a power source and the PPM pad; a pull-down driver electrically connected between a ground and the PPM pad; and a filter coupled to the PPM pad, wherein the communication interface is configured to receive a PPM circuit selecting signal for enabling the pull-down driver or the pull-up driver of the memory die to be switched on.
34. The memory die of claim 33, wherein the PPM circuit further comprises: a pull-up resistor electrically connected between the power source and the PPM pad, wherein the pull-up resistor is connected in series with the pull-up driver.
35. The memory die of claim 34, wherein the PPM circuit further comprises: a pull-down resistor electrically connected between the ground and the PPM pad, wherein the pull-down resistor is connected in series with the pull-down driver, wherein one of the pull-up driver and the pull-down driver is controlled by a control signal corresponding to a peak power operation performed on the memory die.
36. The memory die of claim 33, wherein the PPM circuit further comprises: a pull-up current source electrically connected between the power source and the PPM pad, wherein: the pull-up current source is connected in series with the pull-up driver and is configured to provide constant current; and the pull-up driver is controlled by a control signal corresponding to a peak power operation.
37. The memory die of claim 33, wherein the PPM circuit further comprises: a pull-down current source electrically connected between the ground and the PPM pad, wherein: the pull-down current source is connected in series with the pull-down driver and is configured to provide constant current; and the pull-down driver is controlled by a control signal corresponding to a peak power operation.
38. The memory die of claim 33, wherein the PPM circuit further comprises: an output circuit for outputting a PPM enablement signal, wherein the PPM pad is electrically connected through the filter with a first input terminal of the output circuit, and the output circuit is configured to output the PPM enablement signal according to an electric potential of the PPM pad.
39. The memory die of claim 38, wherein the output circuit comprises: a comparator with a first input terminal electrically connected through the filter to the PPM pad and a second input terminal electrically connected to a reference voltage.
40. The memory die of claim 39, wherein the reference voltage is based on a maximum number of peak power operations allowed for multiple memory dies of a storage system.
41. The memory die of claim 39, wherein the output circuit further comprises: an inverter with an input connected to an output terminal of the comparator, wherein the inverter is configured to output the PPM enablement signal for determining whether to perform a peak power operation.
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